



Project Title

SkyPort: Geographic Simulation for Urban Air Mobility

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REQUIREMENTS ANALYSIS DOCUMENT

1. Introduction

This document has been prepared to present the requirements analysis and technical framework of the SkyPort project. This project offers an innovative solution in the field of urban air mobility. SkyPort is a Geographic Decision Support System (GDSS) model that digitizes manual site selection processes through open-source spatial data and advanced decision-making algorithms. By integrating live open-source spatial data with sophisticated Multi-Criteria Decision Making (MCDM) algorithms over a discrete hexagonal spatial grid, the system delivers mathematically optimized infrastructural insights. With this system is its dual-actor architecture, analysts and passengers no longer operate in separate silos. They share a common platform where infrastructure decisions and travel planning feed into each other.

Throughout the document, the system's core functions, operational limitations, engineering success criteria, and the project's academic objectives are discussed. It is intended to be used as a reference document during the design and implementation phases for all modules in the development process.

1.1. Purpose of the System

The main objective of this project is to design and develop a web-based Geographic Decision Support System (GDSS) leveraging open-source geographic data to identify the optimal location for the establishment of the Vertiport, considering the Urban Air Mobility (UAM) scope in Istanbul, and then to establish a web-based bidirectional system to enable the use of the network for the end-users.

The system also has the ability to analyse multiple criteria such as height (obstacles), transport networks, land use, meteorological conditions, traffic density, socioeconomic factors, and restricted flight zones. With the Multi-Criteria Decision Making (MCDM) algorithm to be developed, it is possible to automatically calculate the "Vertiport Suitability Score" for coordinates that can be set by the user depending on their needs. In such a manner, it is possible to provide a flexible decision support tool for institutions, organizations, and individuals interested in investing in Vertiports. Additionally, it is also possible to save the results obtained from previous analyses.

The end-user (B2C) module of the system directly provides passengers with access to the vertiport network, which has been approved through analysis. Passengers can filter vertiports according to their personal preferences (cost, distance, noise, etc.), select the most suitable point for their travel plans, and add frequently used locations such as home/work to their favourites for a personalized experience.

In conclusion, unlike a standard navigation or map application, this system aims to be a systematic engineering tool for visualizing manual feasibility processes with interactive heatmaps and simulated safe flight routes, and also an end-to-end service portal for integrating passengers into the UAM ecosystem.

1.2. Scope of the System

SkyPort is a two-way (B2B and B2C) interactive platform that helps users plan and experience Urban Air Mobility (UAM) infrastructure specifically in Istanbul. Using different geographic data sources, the system provides data-driven decision support for vertiport site selection, while also making this validated network accessible to end-users.

The system offers the following core capabilities:

- **Globally Scalable Software Architecture:** A general analysis platform using Istanbul as a pilot region but adaptable to any metropolis in terms of its data structure.
- **Full-Stack Geographic Development:** The design and implementation of a **robust spatial** database (PostgreSQL/PostGIS), a Python-based **algorithmic** data processing layer, and a React/Mapbox-based **interactive** graphical user interface.
- **Safe Route Analysis:** Developing a basic routing algorithm that avoids obstacles between selected stations.
- **Data-Driven Location Analysis:** By utilizing spatial indexing (such as the H3 Hexagonal Grid system), this involves the integrated analysis of building height (obstacles), transportation infrastructure, traffic density, and socioeconomic factors, which can be obtained from OpenStreetMap and other open data sources.
- **Dynamic Criterion Weighting (MCDM):** This enables the user to set their own parameters for the Multi-Criteria Decision Making (MCDM – AHP/TOPSIS) algorithm, which will be integrated into the decision support system.

- **Visualization and Simulation:** Providing visualization of analysis results through an interactive heatmap on a Mapbox-based map and simulating safe flight routes without obstacles from one location to another.
- **Analysis Management and Reporting (B2B):** Providing experts with an ability to compare various location alternatives based on their scores, view detailed results of each location analysis, save previous results of location analysis to experts' profiles, and export results.
- **End-User Interaction (B2C):** Providing end users with an ability to view an approved network of vertiports, view results of filtering by their own preferences (cost, distance, noise level, etc.), choose travel destinations, and save frequently visited locations to favourites (Home, Work, etc.).

The system's core user roles are:

1. **Expert User (Urban Planner/Aviation Investor-Analyst):** User will carry out strategic analysis by linking the provided Vertiport infrastructure to urban planning and will report the results. The user will assess the operational safety and commercial potential of potential investment locations, identify the most suitable investment location, and use comparison tools.
2. **Customer/Passenger:** The customer will filter the provided vertiports according to their criteria using the system, select the points for boarding and disembarking for their personal use, and manage their favourite points.

Out of Scope

To define the system's limitations, the following features have been excluded from the project scope:

- Real-time air traffic control, flight tracking, or handling of aircraft.
- Financial ticketing, payment infrastructures (payment gateways), or management of actual commercial flight reservations (The Passengers will only plan their flight using the system and select/filter the points; there is no financial ticketing for them to purchase a flight.)
- Physical hardware, sensor development, in-vehicle software integration.
- Real-time data analysis for cities other than Istanbul

1.3. Objectives and Success Criteria of the Project

The project's success criteria are as follows:

- The delivery of a fully functional web simulator, accessible via a standard web browser, for personal computers, acting as a proof of concept for both professionals (B2B) and passengers (B2C).
- The Multi-Criteria Decision Making (MCDM) algorithm developed must effectively process physical barriers (mania), transportation integration, environmental, and socioeconomic impact criteria to produce an understandable "Vertiport Suitability Score" while ensuring expert users can effectively save the results of their analysis to their profiles and export them in report form.
- Approved vertiport locations should be displayed perfectly on the customer/passenger interface, and it should be possible to filter these locations according to personal preferences

(distance, costs, noise level, etc.), choose boarding and disembarking points, and mark frequently used locations as favourites.

- The proposed three-tier software architecture (React front-end, Python back-end, PostgreSQL/PostGIS database) and actor-specific session management (login/logout) infrastructure should be perfectly integrated, and the system should work without critical errors.
- Suitability results should be displayed in an interactive heatmap format, and safe route suggestions should also be provided to the user between the chosen optimum locations.
- Project should be completed in the proposed timeframe, and all proposed functional and non-functional requirements should be met.
- The developed system should demonstrate environmental and social sensitivity in line with United Nations Sustainable Development Goals 9 (Industry, Innovation, and Infrastructure) and 11 (Sustainable Cities and Communities)

1.4. Definitions, Acronyms, and Abbreviations

- SkyPort: Name of the proposed and developed bidirectional (B2B and B2C) Decision Support System and vertiport network platform.
- RAD (Requirements Analysis Document): Document that describes the scope, goals, and functional specifications of a proposed project.
- UAM (Urban Air Mobility): Technological environment that provides urban air mobility.
- eVTOL (Electric Vertical Take off and Landing): Next-generation aircraft that are electric-powered and allow vertical take off and landing.
- Vertiport: Locations where operational activities of eVTOLs take place, take off and landing and charging of aircraft, and transferring of passengers.
- MCDM (Multi-Criteria Decision Making): Algorithmic structures that identify the most suitable decision alternative based on various criteria (AHP and TOPSIS techniques) in decision-making processes.
- Obstacles: Buildings, towers, and high-voltage lines that impact flight safety of aircraft.
- PostGIS: An extension of PostgreSQL that provides geographic/location-based (spatial) queries in a database environment.
- NFZ (No Fly Zone): Locations where flight is prohibited.
- B2B (Business-to-Business): The platform's expert (management) module focused on analysis and reporting, used by urban planners and aviation investors.

- B2C (Business-to-Consumer): The platform's end-user module serving customers and passengers directly, focusing on filtering, booking scheduling, and favourite management.
- User: any person who uses the system.

1.5. Overview

This Requirements Analysis Document (RAD) is structured to present the technical, architectural, and functional details of the SkyPort bidirectional (B2B and B2C) Decision Support System and platform, and consists of six main sections:

- **Section 1 : Introduction:** This section provides an overview of the major framework of this project and includes a brief overview of the reasons for developing this system, the purpose of this system, scope, success criteria, and major definitions.
- **Section 2: Current System:** This section is intended to analyse the current manual feasibility studies or digital solutions available in the field of Urban Air Mobility (UAM), and it will be made clear which issues are being addressed.
- **Section 3: Proposed System:** This is the major part of this document and includes an overview of functional requirements (Use Cases and Scenarios), non-functional quality attributes, and system models that this system will provide to users, including professionals (B2B) and passengers (B2C).
- **Section 4: Other Analysis Elements:** This section includes a detailed project plan, which includes risks that can be encountered during this project process and possible alternatives.
- **Section 5: Glossary:** Provides comprehensive definitions necessary for the correct interpretation of technical terms, abbreviations, and field-specific concepts used in the document.
- **Section 6: References:** Lists all academic publications, technical documentation, and data sources used in the requirements engineering and initial stages of the project.

2. Current System

Today, Urban Air Mobility (UAM) is a brand-new and developing industry, and therefore, there is no standardized and publicly available integrated software for the selection of the vertiport and the delivery of the network to the end user.

Currently, these studies are carried out by city planners and aviation companies. When carrying out a study for a specific region to check the suitability for a UAM network, 3D data for buildings, transportation network data, and zoning data need to be collected separately from different organizations. Analysis is performed manually with a high margin of error, and the process takes months. Currently available mapping tools, such as Google Maps, can only be used for navigation.

However, the current systems do not have the ability of a Geographic Decision Support System, which can analyse building heights, land use types (noise sensitive areas), and transportation networks simultaneously. The absence of an academic and practical tool that can process complex spatial data and display the results as a heat map simulation is the major weakness of the current system. From the point of view of the users/passengers, the current UAM system does not have a digital platform for passengers, potential customers, to plan their future eVTOL (air taxi) routes and filter and view approved active vertiports according to their own preferences (cost, distance, noise).

The SkyPort system aims to eliminate these limitations. It offers a Geographic Decision Support System that can perform spatial analysis quickly and systematically by automatically integrating different sources of geographic data. The system uses multi-criteria analysis to evaluate the possible locations for a Vertiport, which are then presented to the user as an interactive heat map. At the same

time, the system also makes the network accessible to passengers and changes the existing manual process to a digitized bidirectional process.

3. Proposed System

This section presents how the SkyPort system works, the stakeholders of the system, the actors involved, and the functional features of the system. The overall structure of the system is presented. SkyPort is not just a Geographic Decision Support System (GDS) prototype to identify the most suitable locations for eVTOL (electronic taxi) vertiports within a city. Instead, it is a web based integrated system that allows the public to access the network directly. It is a bidirectional web based integrated system (B2B, B2C) applicable to the city of Istanbul as a pilot city.

3.1. Overview

SkyPort is a bidirectional (B2B and B2C) web based integrated Geographic Decision Support System (GDSS), designed to identify the best locations for vertiport infrastructure under the context of Urban Air Mobility (UAM), and to make the network available to end users.

The SkyPort system is a comprehensive analysis platform that allows users to:

- Select and manage the urban areas to be analysed via the map interface;
- Real-time retrieval of building height (mania), transportation infrastructure, and land use data from open data sources (OSM) and index them on an H3 hexagonal grid architecture,
- Dynamically perform vertiport suitability analysis and compare options using Multi-Criteria Decision Making (MCDM/AHP/TOPSIS),
- Visualize analysis results in interactive heatmaps and save them as reports to expert profiles,
- Compare alternative Vertiport locations,
- Plan safe travel routes between potential Vertiport points and manage their favourite locations,
- Allow passengers to view the vertiport network approved by experts and filter according to their personal priorities such as distance, cost or noise.

The SkyPort system includes various stakeholders and actors who interact with the system and contribute to its development.

Actors:

1. Expert User (Urban Planner / Aviation Investor) - [B2B] This is the main actor for the analysis, calculation, and management (B2B) part of the system. This actor will choose regions on the map, set the weight parameters for the Multi-Criteria Decision Making (MCDM) algorithm according to their strategic or commercial goals. This actor will evaluate the suitability analysis results provided by the system.

2. Passenger - [B2C] This is the actor for the customer portal (B2C) side of the system. This actor does not interact with the analysis algorithm. Instead, they will look at the active vertiport network

that has been analysed and approved by the Expert User on the map. This actor will filter the vertiport based on their travel preferences. This actor will choose the boarding/disembarking points for the vertiport. This actor will add the routes to the favourite list.

Stakeholders:

1. Civil Aviation Authorities (e.g., DGCA): Urban Air Mobility System needs to adhere to the civil aviation regulations and airspace regulations (No Fly Zone - NFZ). Though they are not using the system as a user through the interface, they are the most important stakeholders since the analysis and simulation carried out by the system have to adhere to their safety regulations.

2. External Data Providers (e.g., OpenStreetMap): The Urban Air Mobility System needs accurate geographical information to perform the analysis. Open-Source Platforms such as OpenStreetMap are external stakeholders who provide the necessary data such as building heights (obstacles), transportation network data, etc. This data is fed into the system in the background.

3. Software Development Team: This team is responsible for the architectural design of the SkyPort System, coding the MCDM Algorithm, PostGIS Database Management, etc. This team ensures the sustainability of the project from a PoC (proof of concept) to a product.

The addition of all the stakeholders and actors to the system allows the Urban Air Mobility System to perform the functions of a data-driven decision support system.

3.2. Requirements

This section highlights the basic functions that are expected from the SkyPort system. The requirements established are the processes through which the system processes the geographical information received from the user, transforming the information into a more useful output, which can either be a map or a vertiport availability score. The processes established are used to create a strategic decision support tool for B2B professionals, while for B2C travellers, the processes are used to create a travel guide, thereby highlighting the two-way value proposition of the system. The functions established by the system, as well as the interaction with the user, are highlighted in the "Use Cases" section below.

3.2.1. Use Cases

SHARED USAGE SITUATIONS (B2B & B2C)

UC1: Creating an Account: Priority: High

Users who want to log in for the first time create an account by choosing the type of account that they want to use the system with their own profile. Once the user enters their email and password, the system checks the user's information and creates a new user profile in the system database. The user can then log in to the dashboard according to their type.

UC2: Login: Priority: High

Registered users who want to log in to the system and use their past analytics or their favourite locations log in by entering their email and password. The system checks the user's information from the database and loads the interface according to their type.

UC3: Simulate Flight Route: Priority: Medium

User being a feasibility testing expert or a passenger planning a trip, selects the departure point and arrival point on the map for the flight route between these two points. The user then requests the system to simulate the safest flight route between these two points. The system then searches for any

physical barriers between these points. It also requests the wind speed from the external weather API and compares it with safety limits. If all the requirements are met, the system utilizes pathfinding algorithms to simulate the safest flight route between these points on the map without any chance of a collision. If the requirements are not met, the system does not allow the user to simulate the flight route and send a cancellation warning message.

EXPERT USER (B2B) SITUATIONS

UC4: Select Analysis Region: Priority: High

Expert user (city planner or investor) selects the target region they wish to analyse in Istanbul by using the integrated Mapbox map interface. This selection may be done by drawing a bounding box on the map or by centring a specific coordinate point on a radius basis. Once the user has selected the region, the system will sequentially carry out the following complex operations:

Firstly, the system will retrieve building (obstacle), transportation, and land data for the selected region from the OpenStreetMap (OSM) data repository or PostGIS spatial database. Then, the selected region will be divided into cells by using the Uber H3 Hexagonal Grid data structure. At the same time, the Multi-Criteria Decision Making (MCDM - AHP/TOPSIS) mathematical model algorithm will be executed by the system to calculate a suitability score between 0 and 100 for each cell. Next, the results will be visualized on the map by creating a heatmap in green for high suitability scores or in red for low suitability or risky areas.

UC5: Adjust Criteria Weights: Priority: High

The decision-making expert can change the levels of importance for different analysis criteria, such as accessibility, noise sensitivity, and restrictions in terms of obstructions, through the control panel on the system interface, based on their strategy. Once the expert saves these parameters, the system checks if their sum is exactly 100%, sends these parameters to the system algorithm, and then instantly calculates the results again, presenting them on the screen in the form of a heat map.

UC6: Inspect Cell Details: Priority: Medium

Expert users can click on any hexagonal cell on the screen, where they will be able to see detailed results of the analysis for that specific cell on the screen. The system will present these results in the detail panel in terms of what constitutes the final suitability score for a given cell on the screen, for example, the impact of obstructions, transportation, and environmental factors on the cell's suitability score.

UC7: Compare Candidate Locations: Priority: Medium

Expert user can compare several Vertiport candidate locations that have high suitability scores, as indicated by the system. The system allows the expert user to view the suitability scores and criterion contributions of the selected locations side by side, which makes it easy for them to compare them and establish which one is suitable.

UC8: Save Analysis to Profile: Priority: Low

Expert users save a heat map analysis that they have successfully completed in their profile for later viewing or future work. The system saves the current view of the map, coordinates, and criterion weights used for the analysis to the "Past Analyses" tab of the expert user's database.

UC9: Export Analysis Report Priority: Medium

Expert user requests that the system download the generated suitability map and location information for them to their computer for later use in meetings or presentations internally. The system renders the current view of the map and score information in GeoJSON data format or PDF report format, then allows the expert user to download it.

PASSENGER / CUSTOMER (B2C) SITUATIONS

UC10: View Active Vertiports: Priority: High

Passenger (end-user) views the available vertiport network on the map that has already been previously analysed and approved by experts and added to the system. The system displays the approved (active) vertiport coordinates from the database on the map layer as a marker (pin) that represents the point where the passenger can receive the service.

UC11: Filter Vertiport Network: Priority: High

The passenger further refines his choices by filtering the active vertiports shown on the map based on his personal needs (e.g., closest distance, most cost-effective location, least noise level, etc.). The system compares the input data from the passenger through the interface with the vertiport data from the database and immediately hides the markers that do not match the criteria from the map layer.

UC12: Manage Favourite Locations: Priority: Low

The passenger adds, updates, or deletes his frequently visited locations like home or work, or personal points at vertiports from his "Favourites" list. The system marks these user-selected points in the database and highlights them on the map by a special icon (star) the next time the user opens the map.

3.2.2. Scenarios**UC1-Description:***Table 1*

Overview	
Scenario Title	Create Account
Description	This is the process by which a user (expert or passenger) who is logging in to the system for the first time registers by choosing an account type in order to use the system with their own profile.
Actors and Interfaces	Actors: Expert (B2B), Passenger (B2C) Interfaces: Registration Form Interface (Frontend), Authentication Service (Auth API / Database)
Initial Status and Preconditions	The user should have access to the application's "Register" page. The email address that the user submits in the form should not be previously registered in the system.
Basic Flow	
Step 1: The user selects the type of account, either Expert/B2' or Passenger/B2C.	
Step 2: The user enters the first name, last name, email address, and password, and clicks the Register button.	
Step 3: The system checks the format of the entered values (email format, password strength).	
Step 4: The system checks whether the email address is unique in the database.	
Step 5: The system adds a new user by storing the user's information and the type of user account selected in the database.	
Step 6: The system shows a confirmation message on the screen, and the user is redirected to the dashboard according to the user type.	
Post Condition	
A new user is successfully registered, and the user's session (login status) is activated.	
Alternative Flow(s)	
1a. Registered Email Error Flow	

UC2-Description:*Table 2*

Overview	
Scenario Title	Login
Description	This is the procedure by which a registered user (expert or passenger) uses their email address and password to log into the system to access their past analytics or favourite locations.
Actors and Interfaces	Actors: Expert (B2B), Passenger (B2C) Interfaces: Login Form Interface (Frontend), Authentication Service (Auth API/Database)
Initial Status and Preconditions	The user needs to have access to the application "Log In" page. The user needs to have a previously created active account in the system.

Basic Flow
Step 1: The user submits the form by entering the email address and password.
Step 2: The user clicks the "Log In" button.
Step 3: The system verifies the provided information (email address and password) by accessing the database.
Step 4: The system identifies the verified user type (B2B or B2C) and establishes a session (session/token).
Step 5: The system redirects the user to the main interface (control panel) according to the authorized user type.
Post Condition
The user is now logged in to the system and can access the interface according to their permissions.
Alternative Flow(s)
2a. Incorrect Credential Flow

UC3-Description:

Table 3

Overview	
Scenario Title	Simulate Flight Route
Description	This is the process where the user chooses their departure and arrival points on the map, then asks the system to plot and analyse the safest route for a flight between these two points.
Actors and Interfaces	Actors: Expert (B2B), Passenger (B2C) Interfaces: Map Interface (Mapbox), Route Algorithm Engine (Python), External Weather API, Pathfinding Algorithm (Python)
Initial Status and Preconditions	User must be logged in and have access to Map Interface. At least two approved/candidate vertiport points must be present on the map.
Basic Flow	
	Step 1: The user chooses one of the Vertiport locations provided by the system on the map as their "Departure" and the other as their "Arrival."
	Step 2: The system queries the PostGIS database for spatial data on building heights, no-fly zones, etc., between these two points.
	Step 3: The system queries the external meteorological API for real-time weather and wind speed at these coordinates.
	Step 4: The pathfinding algorithm calculates the shortest route that avoids no-fly zones and other obstacles such as skyscrapers, etc.
	Step 5: The system displays a linear route on the map, calculated by the pathfinding algorithm, as a safe route for a flight.
Post Condition	
	A flight route conforming to safety standards was successfully established between the two points, and flight metrics were presented to the user.
Alternative Flow(s)	
	3a. The system will show a warning if a safe route cannot be established between the selected points due to the density of obstacles.

UC4-Description:

Table 4

Overview	
Scenario Title	Select Analysis Region
Description	This is the process where an expert user selects an urban area of interest for which they would like to carry out a feasibility study through the map interface, and the system extracts the geographic data for that area, displaying the MCDM algorithm's suitability scores on the map. r.
Actors and Interfaces	Actors: Expert (B2B) Interfaces: Map Interface (Mapbox), Data Extraction Service (OSM), MCDM Algorithm Engine (Python), PostGIS Database
Initial Status and Preconditions	The expert user (urban planner/investor) has logged into the system and accessed the map screen. The Mapbox-based map interface should have been loaded successfully.
Basic Flow	
	Step 1: The user opens the integrated map interface provided by Mapbox and activates the "Select Area" tool on the map.
	Step 2: The user selects the area of interest that needs analysis by creating a bounding box or specifying a centre point and radius on the integrated Mapbox map interface.
	Step 3: The user clicks on the "Start Analysis / Confirm" button.
	Step 4: The system checks if the coordinates of the selected area are within the boundaries of Istanbul.
	Step 5: The system stores the area of interest that was selected by the user and prepares it for data retrieval.

Step 6: The system retrieves the coordinates of the area of interest that was selected by the user, sends a request to OpenStreetMap services, and retrieves building (obstacle), transportation, and terrain data for that area of interest.

Step 7: The system stores the data that was obtained in Step 6 into the PostGIS database for processing in the "Multi-Criteria Decision Making" algorithm, notifies the user that data retrieval was successful, and is ready for the "Simulation" phase.

Step 8: The system divides the target region into thousands of discrete hexagonal cells using an Uber H3 Hexagonal Grid structure.

Step 9: Using the collected data, the system simultaneously runs the Multi-Criteria Decision Making (MCDM - AHP/TOPSIS) mathematical model and calculates a "Vertiport Suitability Score" between 0 and 100 for each cell.

Step 10: The system visualizes these calculated scores as an interactive heatmap on the map, using green for high suitability and red for low suitability or risky areas.

Post Condition

Spatial data for the selected region has been successfully processed, and a heat map simulation has been generated on the screen. The user is ready for a detailed examination.

Alternative Flow(s)

4a. Data Source Unavailable (API Error)

4b. Selected Region Too Large (Performance Limit)

4c. If the selected area is outside the Istanbul city limits, the system displays a warning message to the user and prompts them to select a new area.

UC5-Description:

Table 5

Overview	
Scenario Title	Adjust Criteria Weights
Description	This is the procedure by which the decision-making expert adjusts the levels of importance for the analysis criteria (transportation, noise, obstacles, etc.) according to its own strategy, as well as the results of the analysis.
Actors and Interfaces	Actors: Expert(B2B) Interfaces: Criterion Control Panel (Sliders), MCDM Algorithm Engine
Initial Status and Preconditions	The user must have successfully completed UC4 and a heatmap must have been generated on the screen.
Basic Flow	
Step 1: The user invokes the "Set Criteria Weights" panel in the map interface.	
Step 2: The system presents the user with the current criteria and their current weights in percentage form via sliders.	
Step 3: The user changes the weights for analysis criteria such as accessibility, noise sensitivity, and obstruction restrictions via the control panel in the system interface.	
Step 4: The user clicks the "Save Values / Recalculate" button.	
Step 5: The system checks whether the sum of the weights that were just input by the user is equal to 100%.	
Step 6: The system sends the new parameters to the background decision engine for renormalization of the AHP matrices.	
Step 7: The system updates the colour distribution in the heatmap in the map based on the newly calculated scores.	
Post Condition	
The analysis results have been successfully updated according to the strategic weights entered by the user.	
Alternative Flow(s)	
5a. If the sum of the criterion weights does not equal 100%, the system will show a warning.	

UC6-Description:

Table 6

Overview	
Scenario Title	Inspect Cell Details
Description	This is the process where an expert user chooses a particular cell of the heat map and analyses in detail the criterion contributions that constitute the Vertiport suitability score calculated for that particular location.
Actors and Interfaces	Actors: Expert (B2B) Interfaces: Map Interface, Detail Analysis Panel

Initial Status and Preconditions	A calculated and visualized heat map is displayed on the screen.
Basic Flow	
Step 1: The user clicks on any hexagonal cell on the heat map that they want to look at in detail. Step 2: The system will query the database for the data on the geographic data and calculated score breakdowns for the clicked cell. Step 3: The system will bring up the "Cell Detail Panel" on the side or bottom of the screen. Step 4: The system will display the calculated suitability score for the clicked cell, as well as the criteria that make up the suitability score for the cell (mania effect, transportation integration, environmental factors, etc.) in the panel.	
Post Condition	
The user will have a clear understanding of the reasons for the suitability of a particular location.	
Alternative Flow(s)	
6a. If the detailed data for the clicked cell is not available in the database, the system will show an error message.	

UC7-Description:

Table 7

Overview	
Scenario Title	Compare Candidate Locations
Description	This is the process where an expert user selects multiple high-scoring vertiport candidate locations suggested by the system and compares the metrics of these regions side-by-side.
Actors and Interfaces	Actors: Expert (B2B) Interfaces: Map Interface, Comparison Mode (Dashboard)
Initial Status and Preconditions	A successful heat map must be displayed on the screen and the user must select at least two different candidate cells for comparison.
Basic Flow	
Step 1: The user selects at least two different candidate locations (cells) that have high suitability scores on the heat map. Step 2: The user clicks on the "Compare Candidates" button in the tools panel. Step 3: The system retrieves all the suitability scores and criterion contributions of these locations (cost, noise, etc.) from the database. Step 4: The system opens a full-screen comparison window (modal) in the user interface. Step 5: The system assists the user in making a decision by displaying the suitability scores and criterion contributions of these locations side by side in a comparison table or graph.	
Post Condition	
A comparative analysis of the candidate locations that have been selected is displayed on the screen, aiding the decision-making process	
Alternative Flow(s)	
7a: In case the user selects only one location, no comparison process will be initiated by the system.	

UC8-Description:

Table 8

Overview	
Scenario Title	Save Analysis to Profile
Description	This is the process by which an expert user saves a successfully completed heat map analysis to their profile for later review or further work.
Actors and Interfaces	Actors: Expert (B2B) Interfaces: Map Interface, Database Registration Service
Initial Status and Preconditions	The user must be logged in and a successfully generated heat map analysis must be displayed on the screen.
Basic Flow	
Step 1: The user clicks on the "Save Analysis" button on the system's interface. Step 2: The user provides a name for the analysis (e.g., "Kadıköy 2026 UAM Plan") in the dialog box that appears and clicks on "OK." Step 3: The system creates a data package containing the current map view, selected coordinate boundaries, and AHP criterion weights used for the analysis. Step 4: The system writes this data package into the user's "History" tab in the database in a secure manner. Step 5: The system displays a message for the user saying "Analysis successfully saved to your profile."	
Post Condition	

The analysis data has been securely saved to the user's profile and can be reloaded in subsequent sessions.
Alternative Flow(s)
8a. If the server or database connection is interrupted during the saving process, the system stops the process.

UC9-Description:

Table 9

Overview	
Scenario Title	Export Analysis Report
Description	This is the process where an expert user downloads the suitability map and location metrics generated by the system to their computer for presentation at internal meetings
Actors and Interfaces	Actors: Expert (B2B) Interfaces: Export API, Report Generator (Render Engine)
Initial Status and Preconditions	A successfully calculated heat map and/or reviewed detail panels must be open on the screen.
Basic Flow	
Step 1: The user clicks on the "Export" button on the interface.	
Step 2: The user selects the report format (Visual PDF Report or Data-based GeoJSON) from the menu that opens.	
Step 3: The system will render the existing map image and scoreboards in the background in the chosen format.	
Step 4: The system will generate a download link for the newly generated file and initiate a download to the user's device.	
Step 5: The system will show a message saying "Report successfully exported" on the interface.	
Post Condition	
The analysis results are downloaded successfully to the user's device in a file format of their choice.	
Alternative Flow(s)	
9a. If the system times out while generating a very large map as a PDF, the process stops and a warning is given to the user.	

UC10-Description:

Table 10

Overview	
Scenario Title	View Active Vertiports
Description	This is the process by which the passenger (end user) views the available vertiport network, previously analysed and approved by experts and added to the system, on a map.
Actors and Interfaces	Actors: Passenger (End User) Interfaces: Customer Map Interface (Frontend), Database Query Service
Initial Status and Preconditions	The passenger must be logged into the system with the "Customer" role.
Basic Flow	
Step 1: The user clicks the "Open Map" or "Show Active Points" button on the customer dashboard interface.	
Step 2: The system queries the approved (status: active) vertiport coordinates and basic information (name, status) from the database.	
Step 3: The system draws the coordinates as interactive markers on the passenger map layer.	
Step 4: The user clicks on any marker on the map to view the name and general status of that vertiport.	
Post Condition	
The passenger has successfully viewed all active vertiport networks available for service on the map.	
Alternative Flow(s)	
10a. If there are no active vertiports yet confirmed in the database, the system opens the map blank and provides information.	

UC11-Description:

Table 11

Overview

Scenario Title	Filter Vertiport Network
Description	This is the process by which the passenger narrows down their options by filtering the active vertiports listed on the map according to their personal needs (closest distance, cheapest location, lowest noise, etc.).
Actors and Interfaces	Actors: Passenger (B2C) Interfaces: Filtering Menu, Map Interface
Initial Status and Preconditions	The user must have completed the previous step, UC10, and the active vertiport pins must be shown on the screen.
Basic Flow	
Step 1: The user opens the filtering menu on the interface and enters the preferred parameters (e.g., "Maximum 5km distance," "Low Noise Area,").	
Step 2: The user clicks the "Apply" button.	
Step 3: The system compares the parameters entered by the user with the properties of the vertiports shown on the map, as indicated by the JSON data.	
Step 4: The system temporarily hides the vertiport markers that do not correspond to the user's requirements from the map.	
Step 5: The system shows the number of filtered results ("3 vertiports found matching your criteria").	
Post Condition	
The map view has been updated to show vertiports that match the passenger's personal preferences.	
Alternative Flow(s)	
11a. If no vertiports remain that match the selected strict filters, the system hides all markers and issues a warning.	

UC12-Description:

Table 12

Overview	
Scenario Title	Manage Favourite Locations
Description	This is the process of adding, editing, or deleting frequently visited locations, such as home or work, or preferred vertiport points, to a "Favourites" list specific to their profile.
Actors and Interfaces	Actors: Passenger (B2C) Interfaces: Map Interface, User Profile (Database)
Initial Status and Preconditions	The user must be logged in with their own session (B2C), and a vertiport point on the map must be selected.
Basic Flow	
Step 1: The user clicks the "Add to Favourites" (Star) icon on the vertiport information card they clicked on the map.	
Step 2: The system will match the ID of the selected vertiport with the current user profile in the database and write it into the "Favourites" table.	
The system will fill the "Star" icon on the interface, and the message "Added to Favourites" will be shown on the screen.	
Step 4: On the next opening of the map, the system will highlight this favourite point with a special icon, different from the standard markers.	
Post Condition	
The relevant location will be saved in the passenger's personal profile and will be included in a quick access list (favourites).	

3.3. Nonfunctional Requirements

This section defines the non-functional constraints and quality attributes of the SkyPort Geographic Decision Support System that determine the system's architecture, quality, and societal impact.

- **Usability:** The system needs to be able to convey complex geographical data (such as building heights, transportation networks, terrain types) in an understandable way to urban planners and aviation analysts. To achieve this, the "Vertiport Suitability Score" calculated by the system will be presented to the user as a heatmap format via the user interface (using React and

Mapbox). The "Vertiport Suitability Score" is not just a number; it also needs to be such that when the user clicks on any cell on the heatmap, they can view the sub-components of the suitability score (i.e., the reason for the suitability score). The passenger interface needs to be such that a normal web user can plan their trip without any navigation assistance.

- **Reliability:** The system is dependent on the OpenStreetMap (OSM) data for map analysis. In case there is a problem in the data provided by this external source for a region (e.g., the height of the buildings is not provided for a region), the system must not crash; rather, it must display a warning message for the user regarding the problem and must continue working in a "partially reliable" mode based on the data that is available (e.g., transportation data and land use data). In cases like the total weight exceeding 100% in the decision-making algorithm, the system must display error messages for the user rather than crashing.
- **Performance:** The performance of the feasibility studies that might take months for urban planners and aviation companies to do manually can be improved by the system's design as a "Geographic Decision Support System." The performance of the spatial queries executed by PostgreSQL/PostGIS and the Multi-Criteria Decision Making (MCDM) algorithm in the Python backend should be able to perform the required calculations and create a heat map for a standard-sized district or neighbourhood selected by the user in a matter of minutes. The focus here is the efficient execution of complex geographic queries by PostGIS rather than the results being returned in milliseconds. The performance in terms of panning and zooming the map should be smooth and should not cause the user to experience lag.
- **Supportability:** The parameters of the Multi-Criteria Decision Making (MCDM) algorithm used in this project (e.g., proximity to the metro = 40%, noise sensitivity = 30%) should not be hardcoded into the code. This is because this is an academic and experimental project, and these parameters should be easily changed in the background depending on various scenarios of Urban Air Mobility (UAM).
- **Implementation:** The project must be developed according to a strict three-tier software architecture. Instead of standard relational databases, PostgreSQL/PostGIS, capable of processing spatial data, will be used in the database layer. The backend will be written in Python, and the frontend in React.js. As per course guidelines, ready-made web templates will not be used for frontend development; custom components will be coded.
- **Interface :** The system must integrate with external web services to programmatically collect and visualize data. The backend will establish API integration with OpenStreetMap services

(e.g., OSMnx) to pull real-world geographic data. The frontend will use the Mapbox Web Services API to process and visualize the generated geographic results (GeoJSON, etc.). The system will not open a commercial API to the outside world but will heavily utilize external data providers. Management of expert and passenger profiles will be provided through a secure login (Authentication) service.

- **Packaging:** The final packaging and delivery of the project should be at the end of the semester. The project is a type of academic research prototype (Proof of Concept), and the final product should be a fully functional web simulator that can be executed in a standard web browser and a local or cloud-based server. The source code for the product should be submitted in its entirety to the academic evaluation committee.
- **Legal :** The source of all the raw geographic data used in the project will be OpenStreetMap. Hence, the system must follow the Open Database License requirements of OpenStreetMap. Legal attributions that indicate that the data used by the application is owned by OpenStreetMap and Mapbox must be clearly indicated on the user interface (the map).
- **Other Constraints (Social, Ethical, Environmental and Business Impacts):**
 - **Sustainability:** The algorithm for the project will be designed based on the United Nations' Sustainable Development Goals (Goal 9-Infrastructure, Goal 11-Sustainable Cities). Due to ethical and health considerations, the algorithm must consider the noise pollution that flying taxis (eVTOL) generate. Land usage must be considered by the algorithm, which must heavily penalize (i.e., assign a low score) the construction of a Vertiport on residential areas like houses, hospitals, and schools, while rewarding those on highways or business premises.
 - **Business and Employment Impact:** The goal of SkyPort is not to eliminate human jobs by automating existing manual processes for feasibility analysis but to provide an up-to-date Decision Support Tool for urban planners and aviation experts to take better decisions. The technology also enables the creation of new fields of expertise for the UAM industry, such as Data-Driven Aviation Planning.

3.4. System Models

This section contains the analysis models of the SkyPort system. The functional structure of the system is visualized with use case diagrams, the data and class structure with object models, and the interaction between components with dynamic models. Additionally, mock-ups showing end-user interaction are presented.

3.4.1. Use Case Model

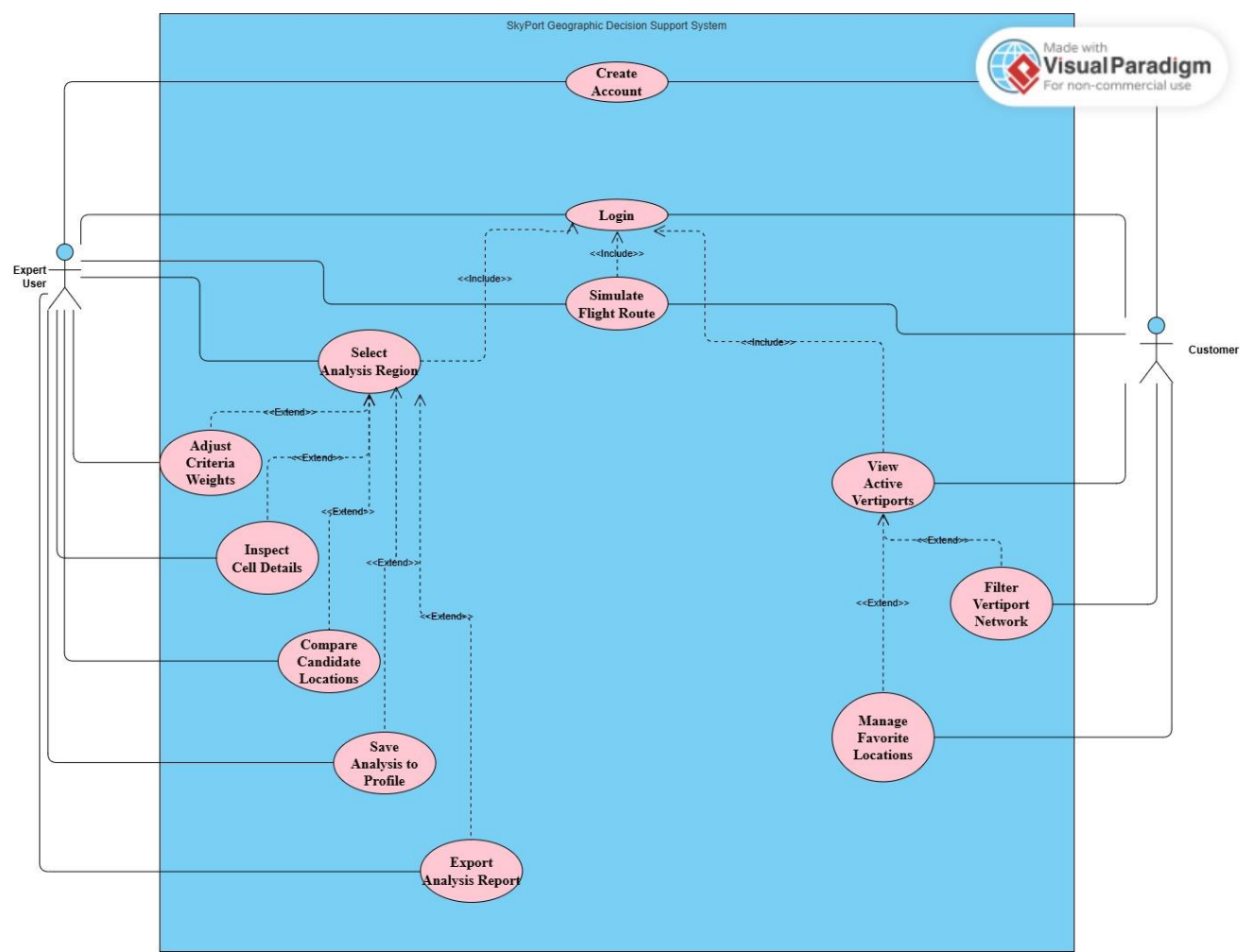


Figure 1

3.4.2. Object Model

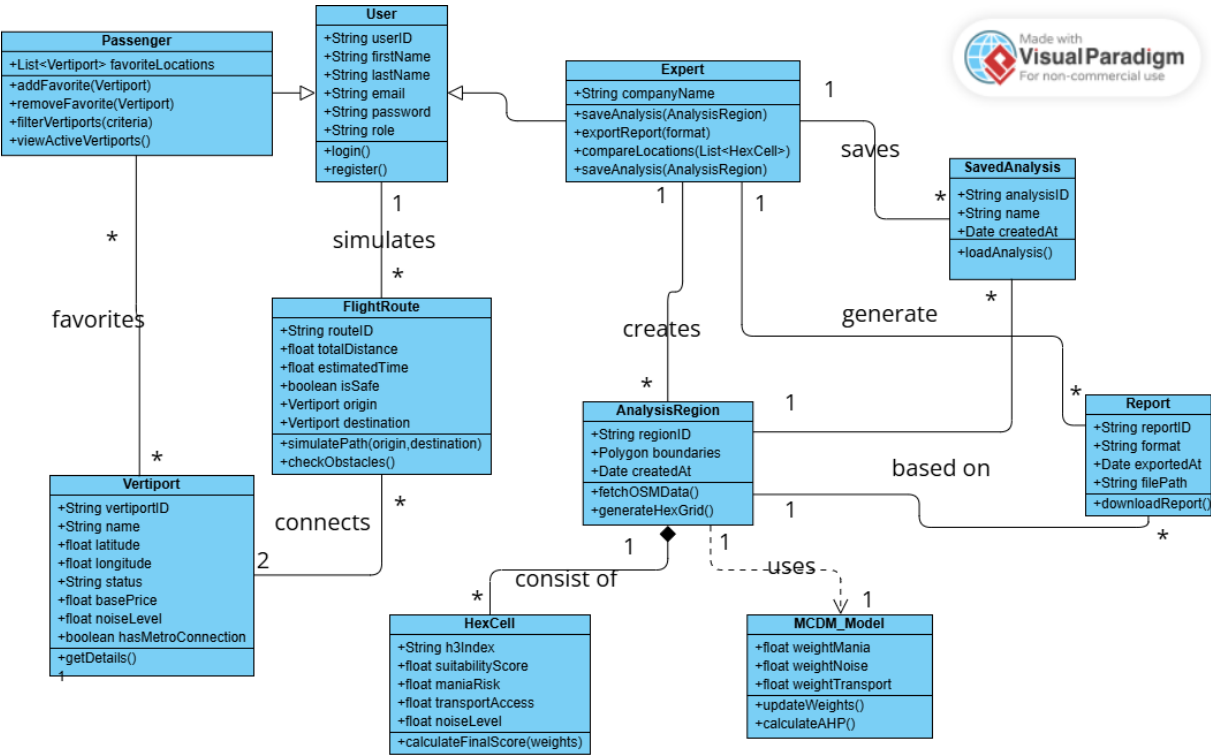


Figure 2

3.4.3. Dynamic Model

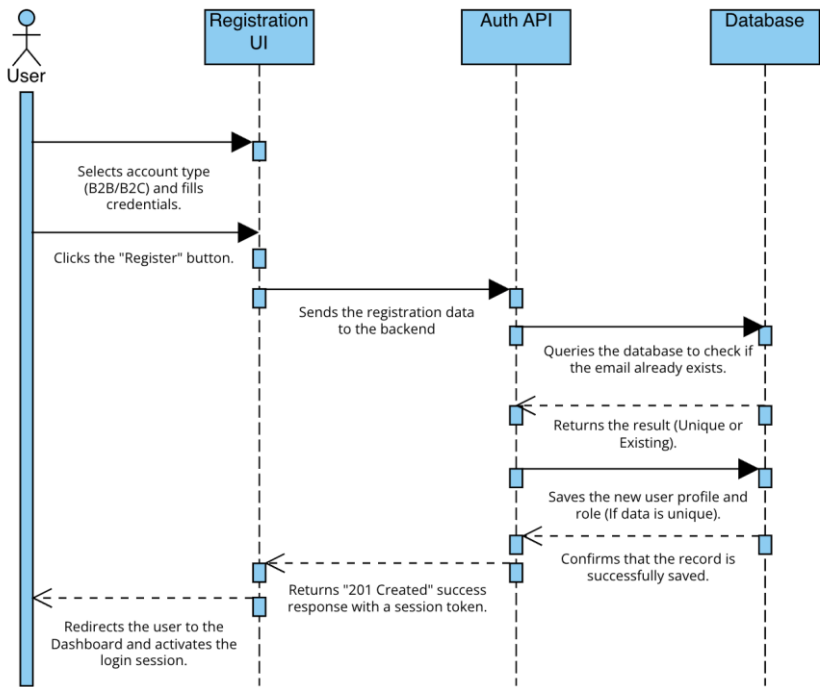


Figure 3: Scenario 1

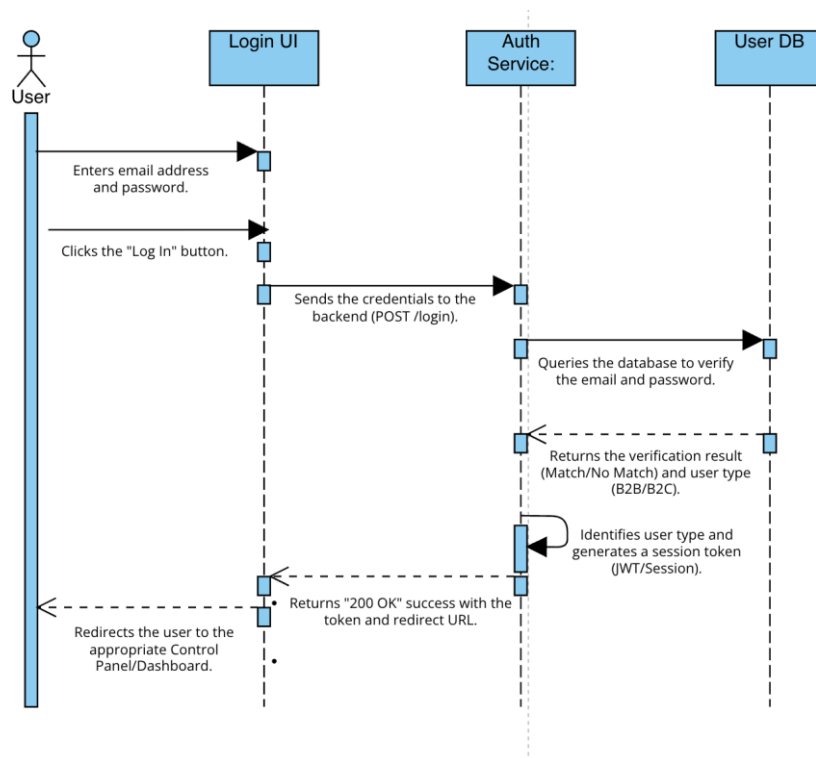


Figure 4: Scenario 2

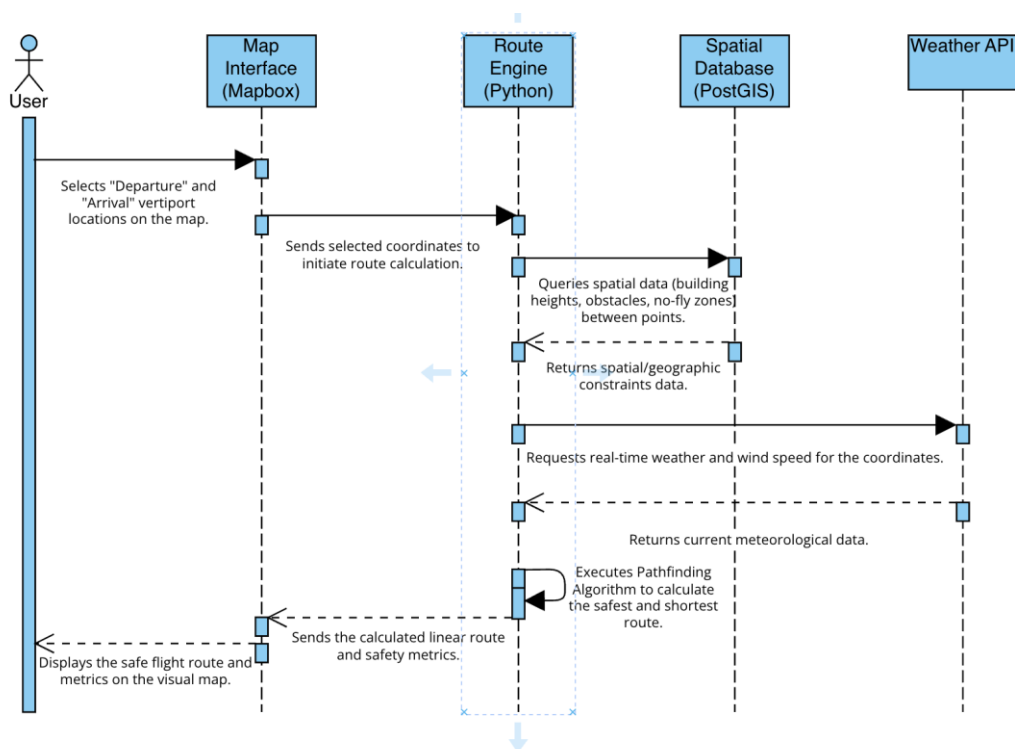


Figure 5: Scenario 3

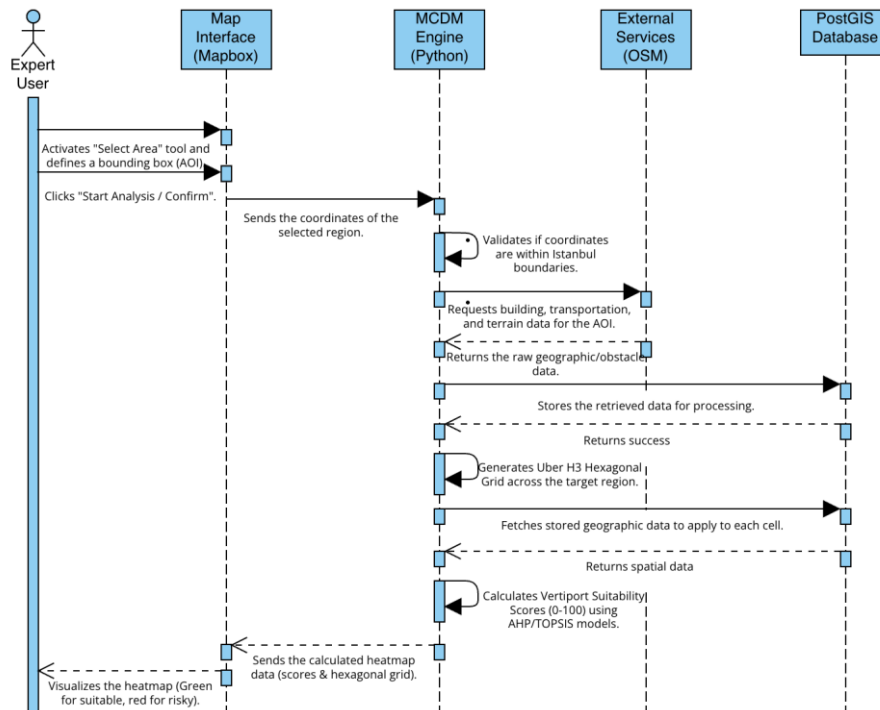


Figure 6: Scenario 4

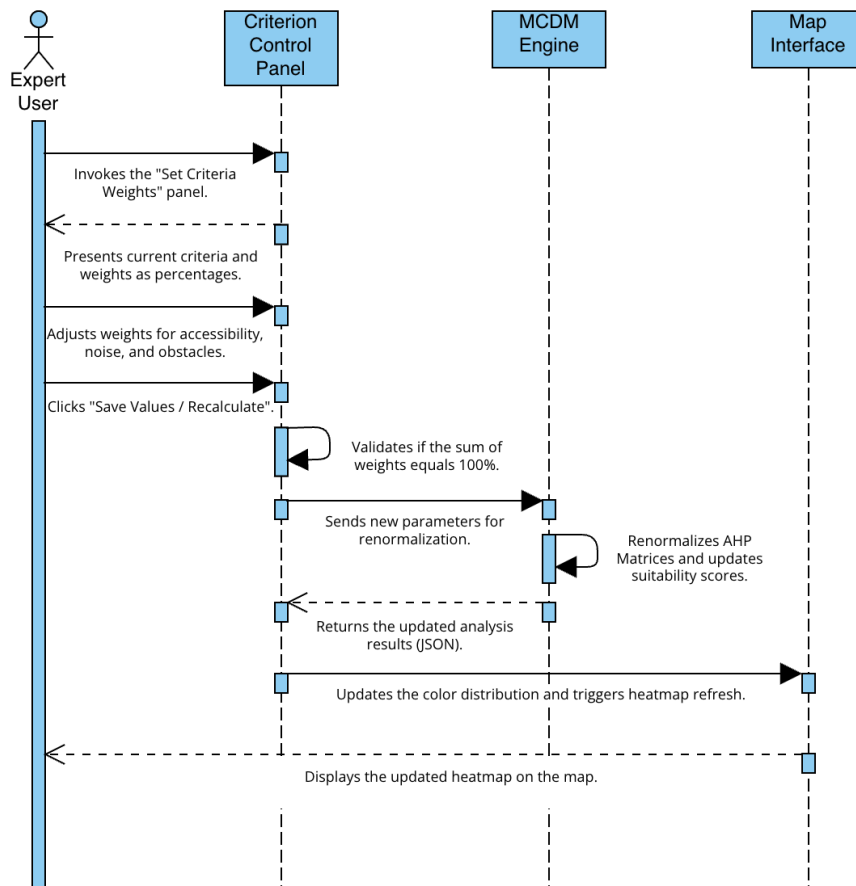


Figure 7: Scenario 5

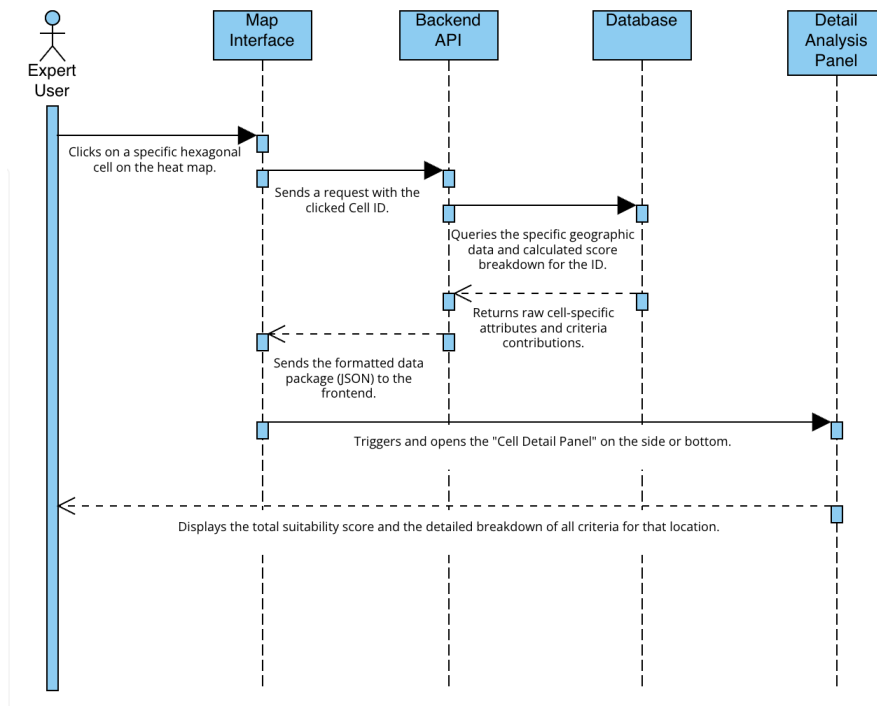


Figure 8: Scenario 6

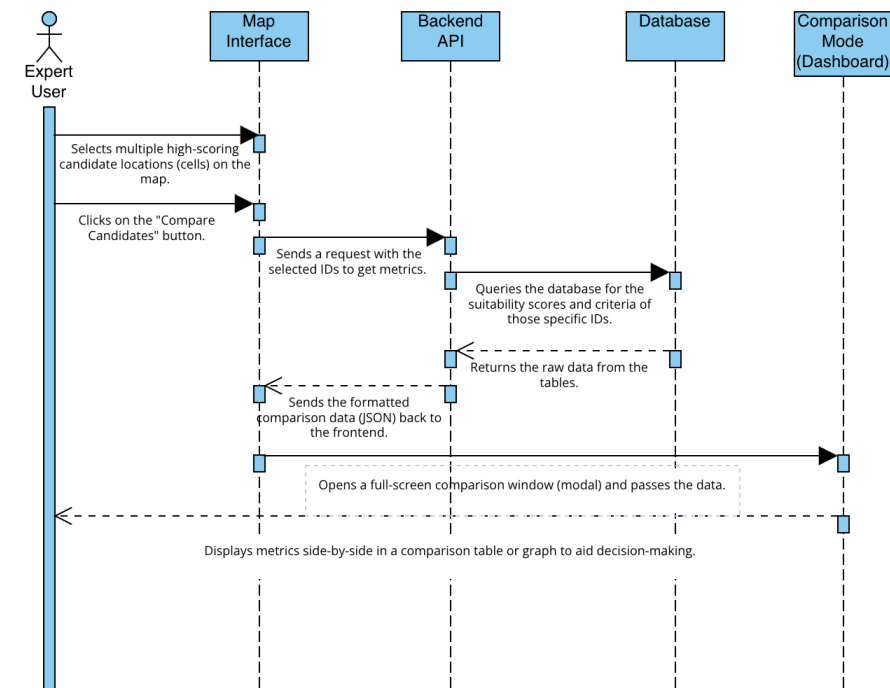


Figure 9: Scenario 7

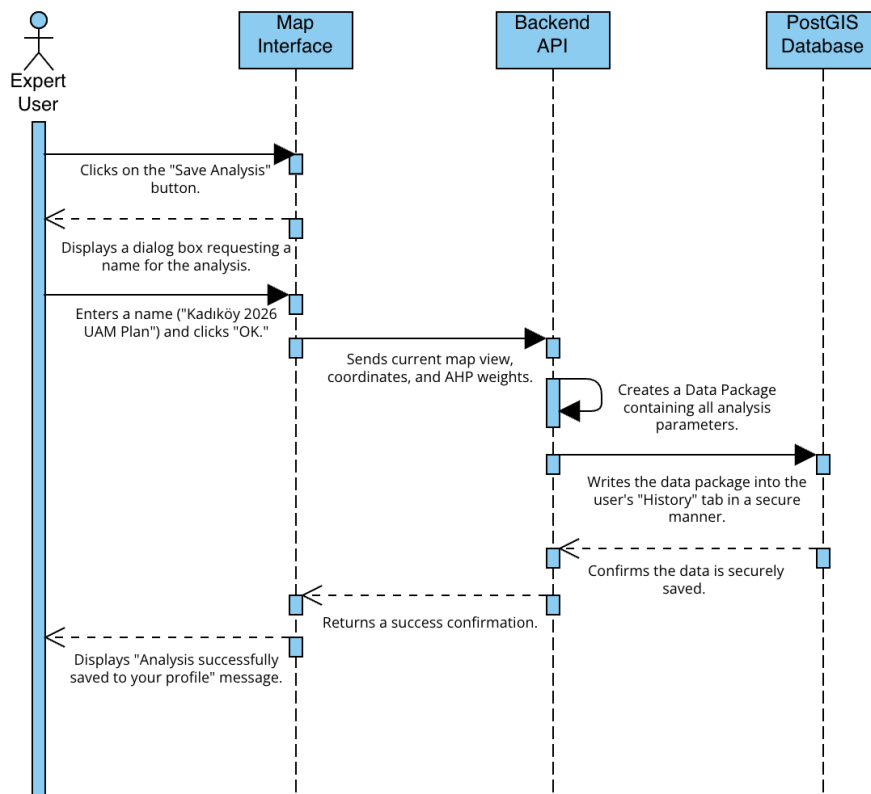


Figure 10: Scenario 8

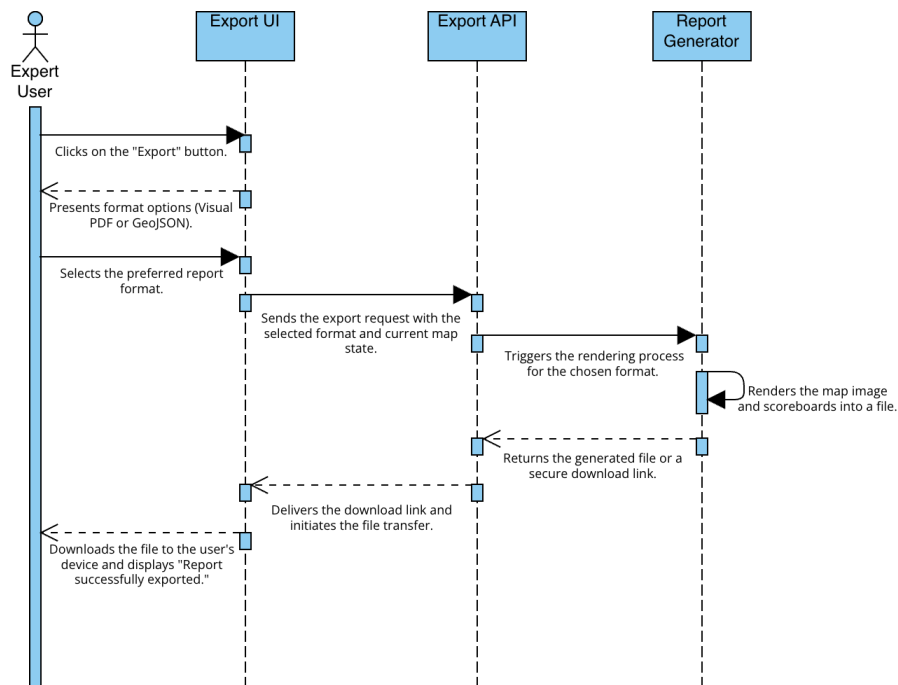


Figure 11: Scenario 9

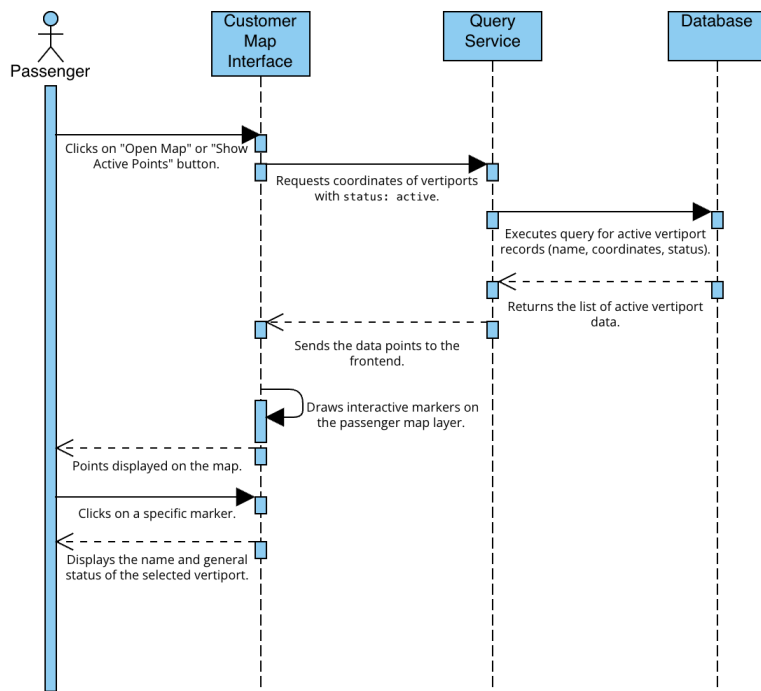


Figure 12: Scenario 10

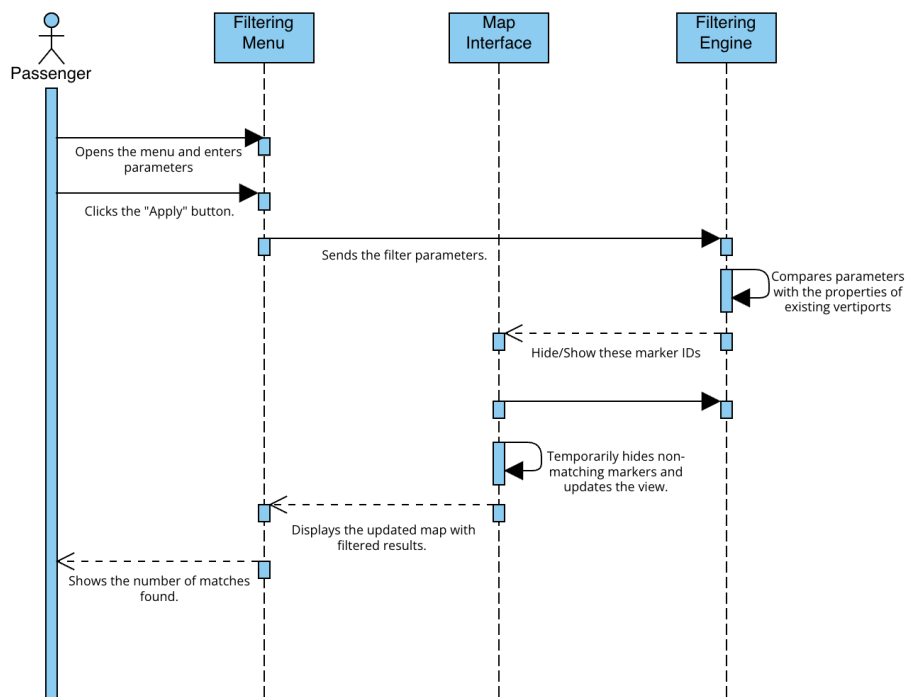


Figure 13: Scenario 11

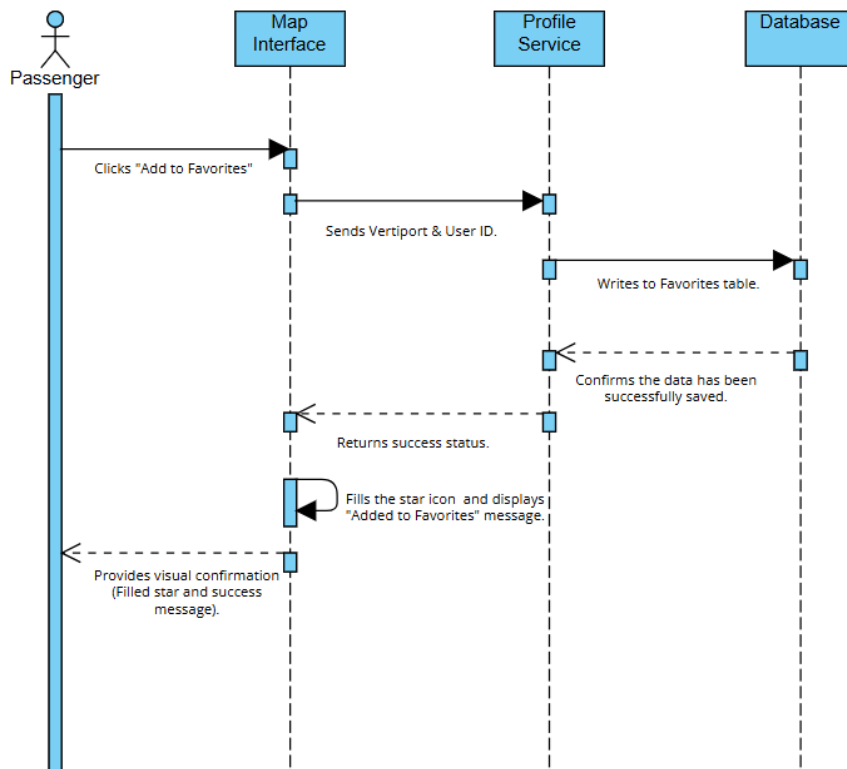


Figure 14: Scenario 12

3.4.4. User Interface—Navigational Paths and Screen Mock-ups

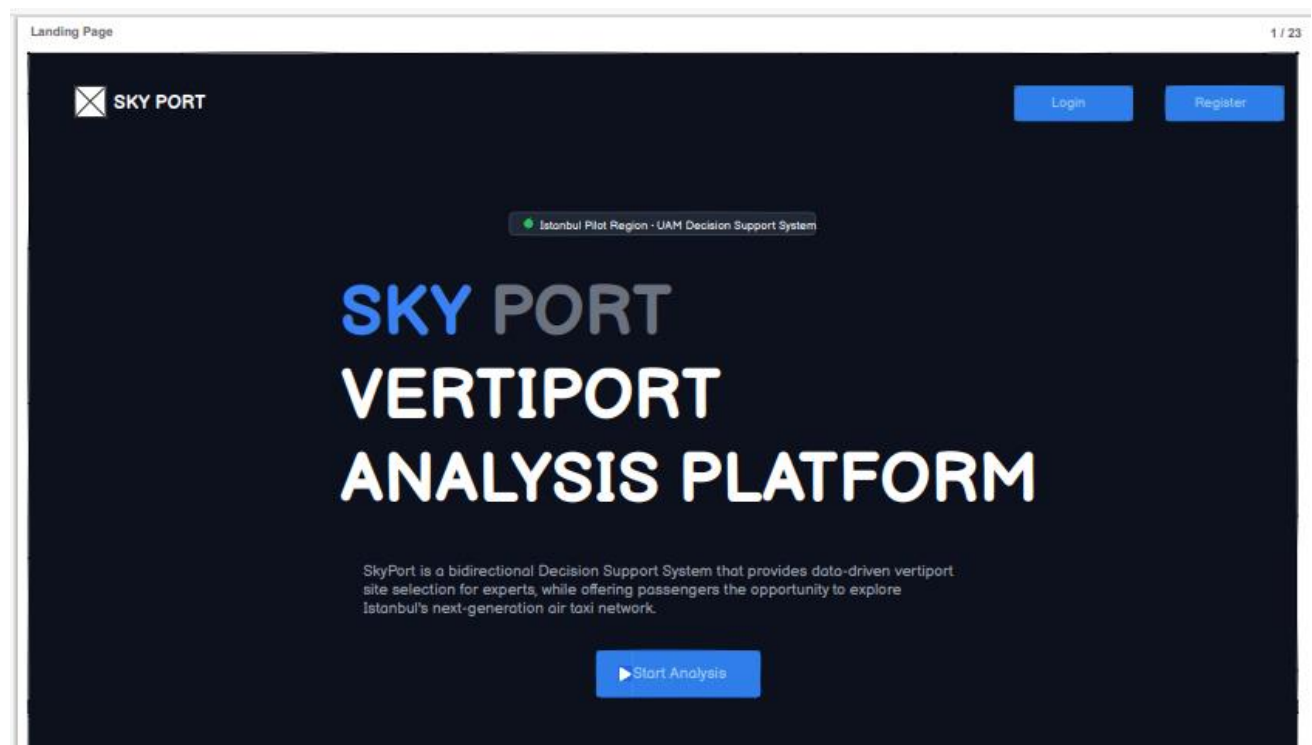


Figure 15

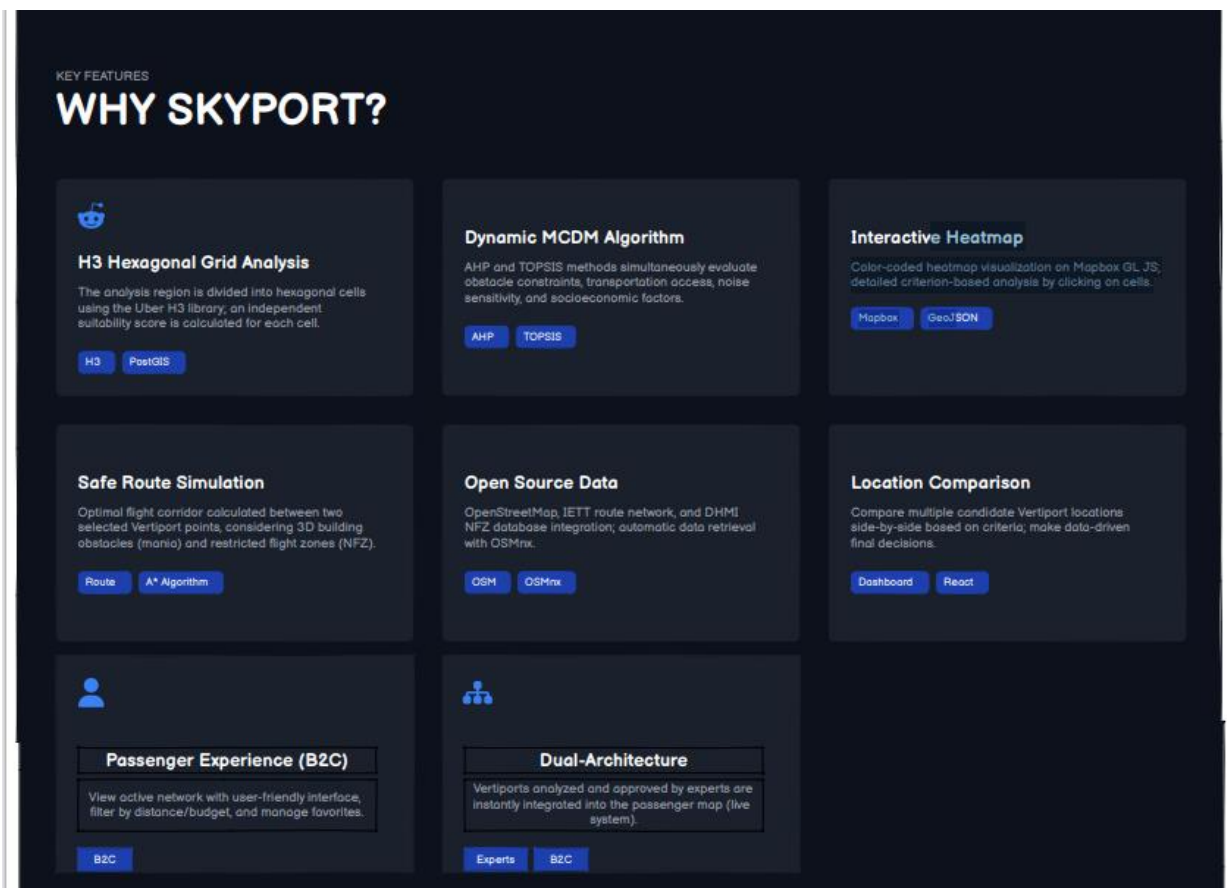


Figure 16

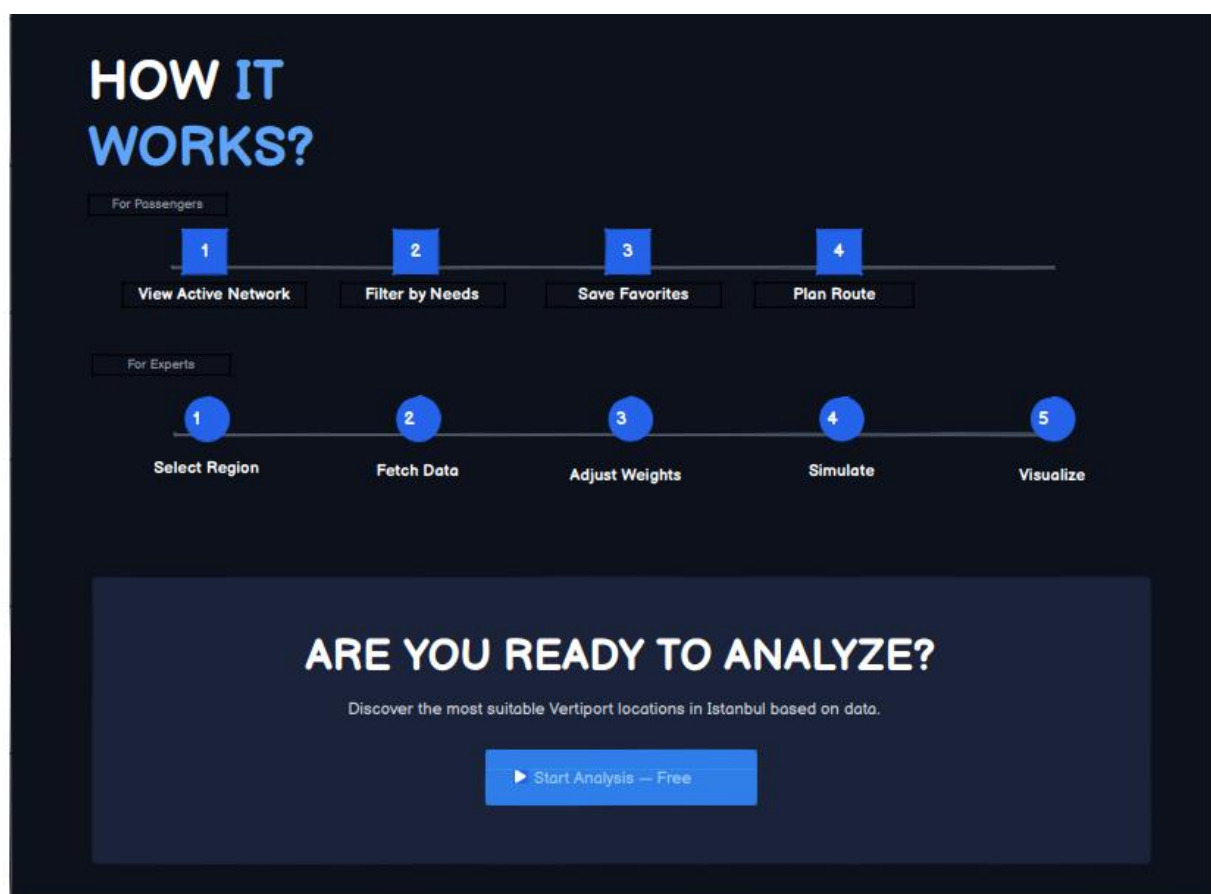


Figure 17

SKYPORT

← → ↻ https://skyport.com

Create New Account

Please select your account type and enter your information.

 Expert (B2B)  Passenger (B2C)

First Name Last Name

Email Address

Create Password

Register

Figure 18

SKYPORT

← → ↻ https://skyport.com

skyport.com/login

Welcome to SkyPort

Log in to your account to continue.

Email Address

expert@skyport.com

Password

.....

Log In to System

Don't have an account? Register (UC1)

Future Transportation Network

We are building Istanbul's air mobility infrastructure with data.

Figure 19

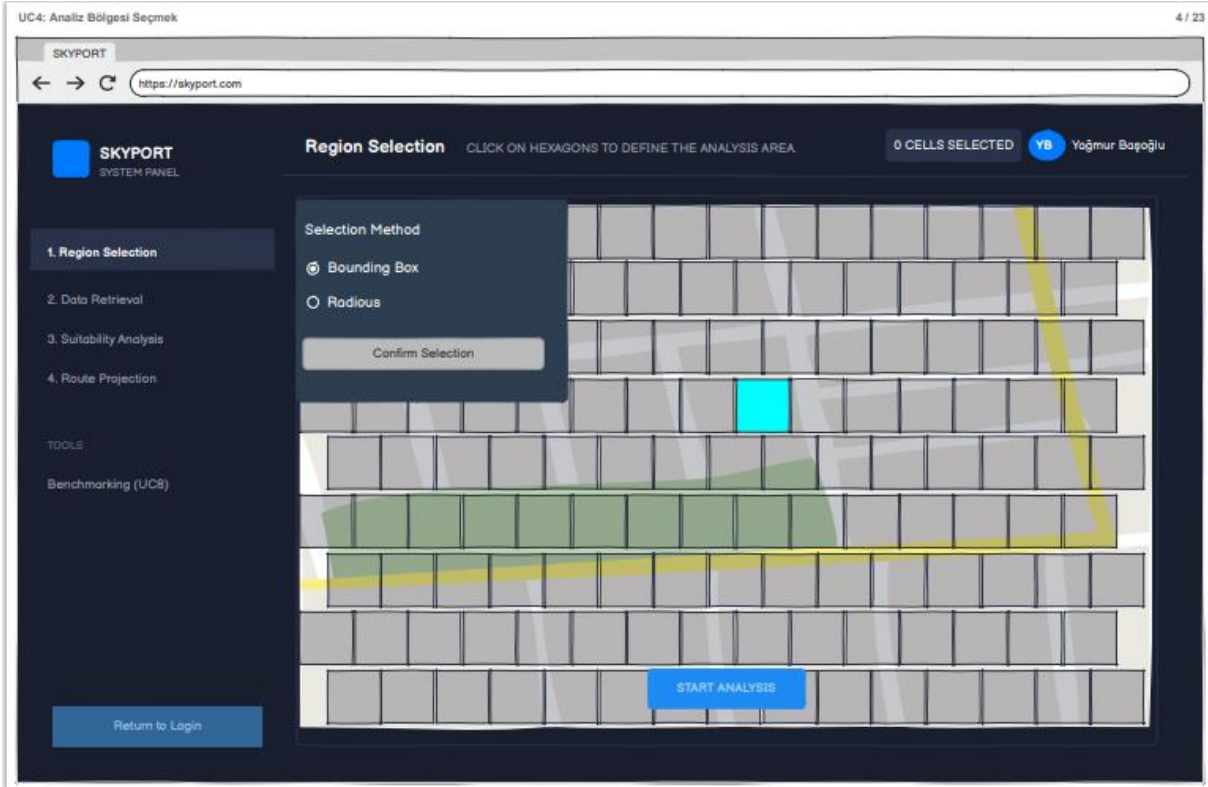


Figure 20

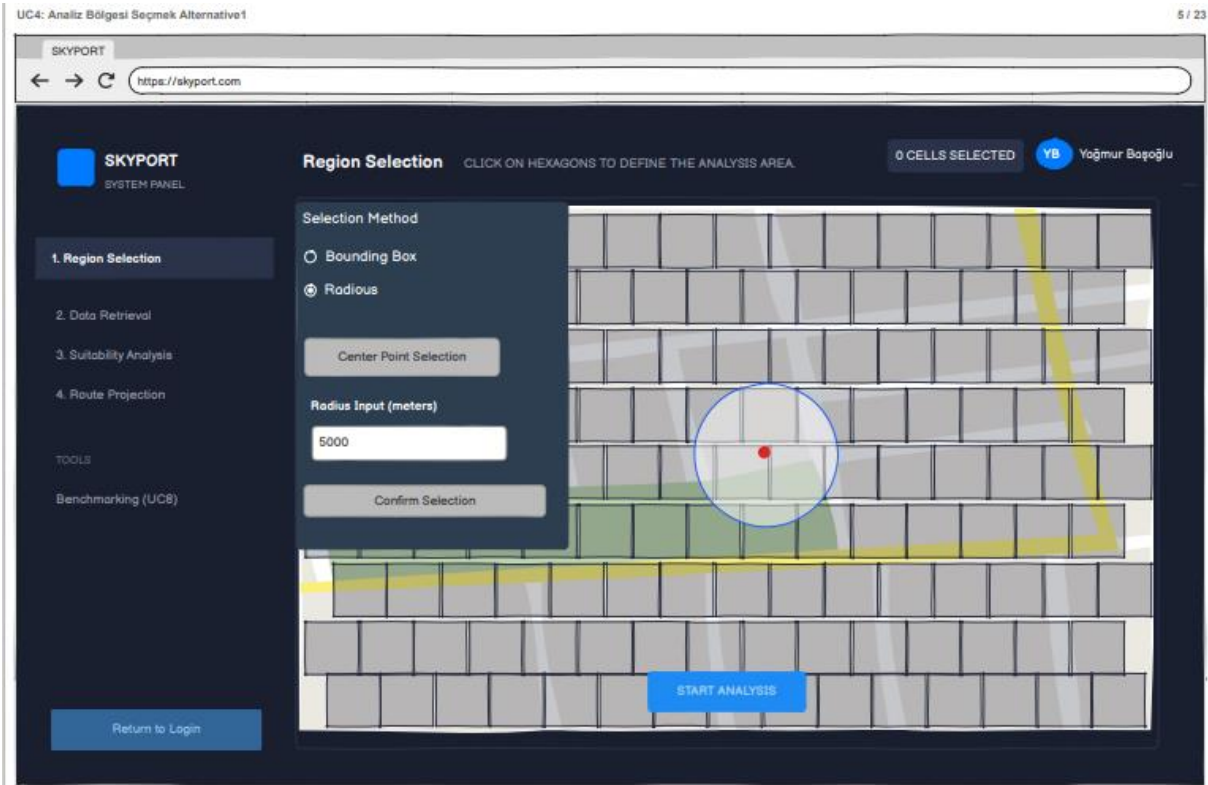


Figure 21

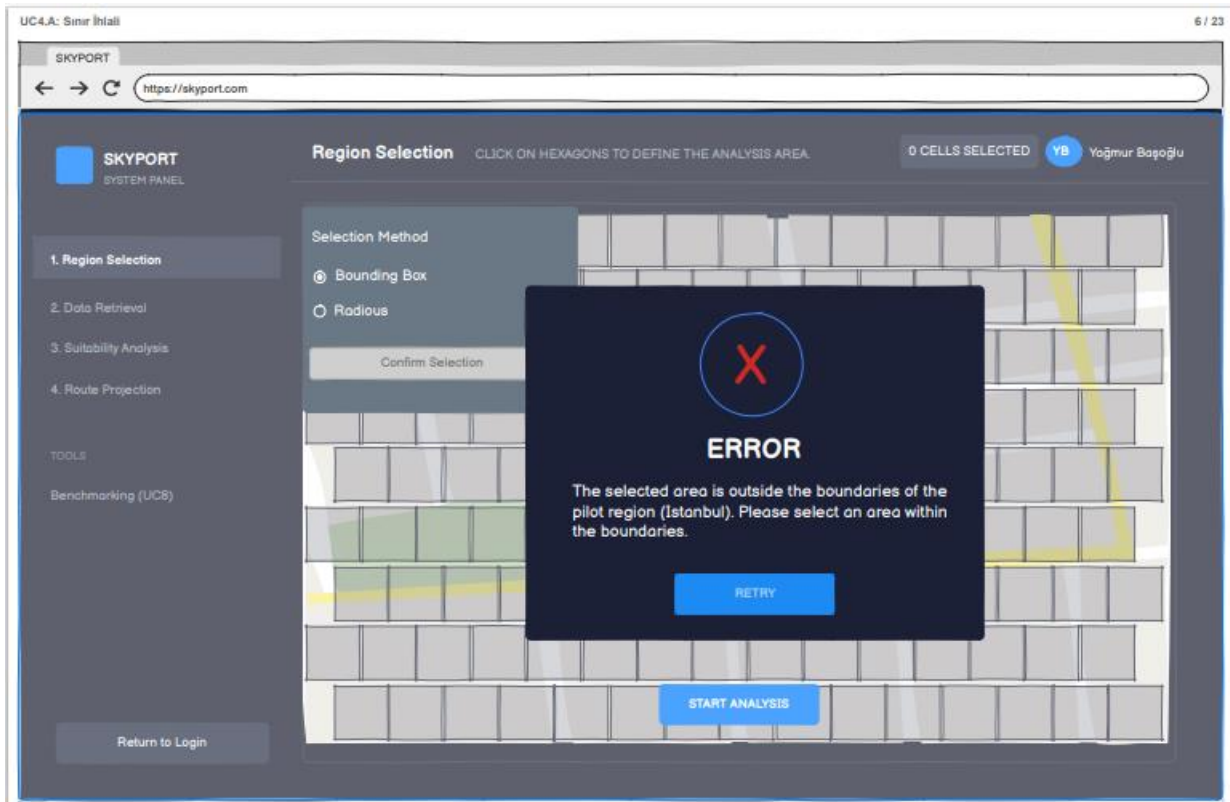


Figure 22

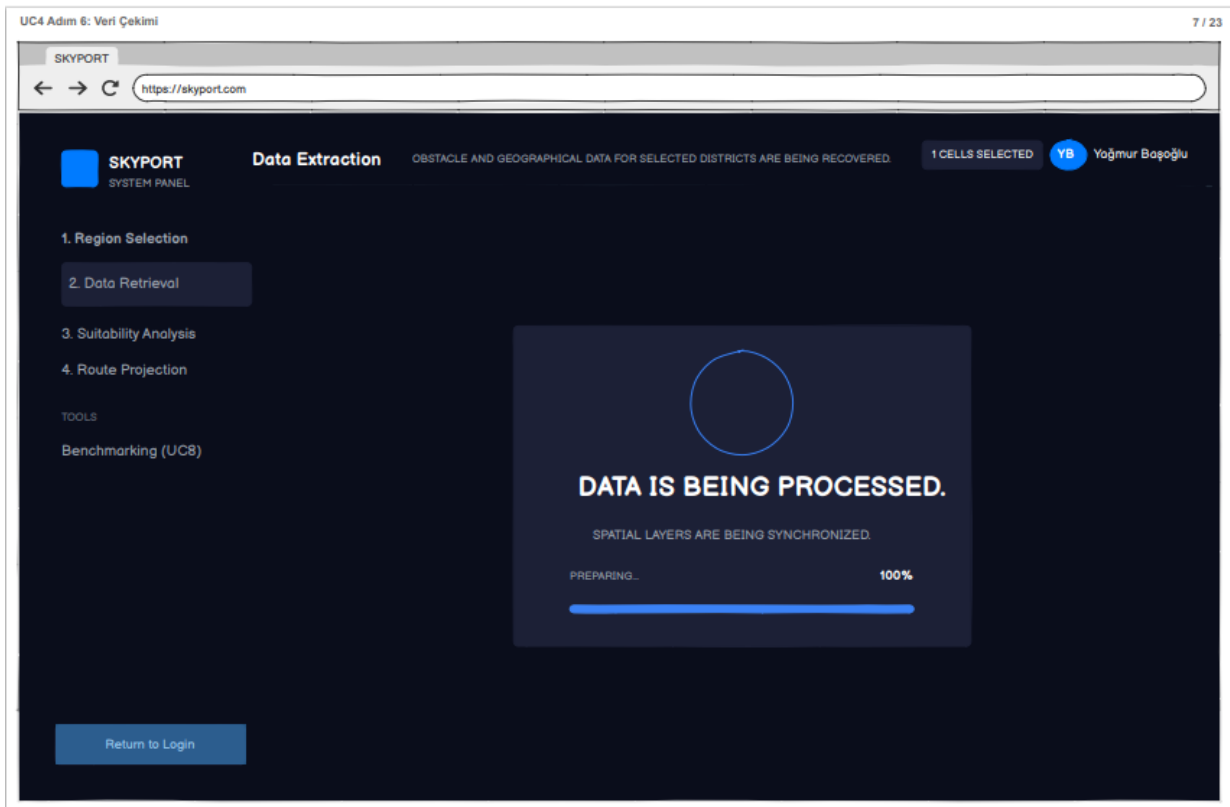


Figure 23

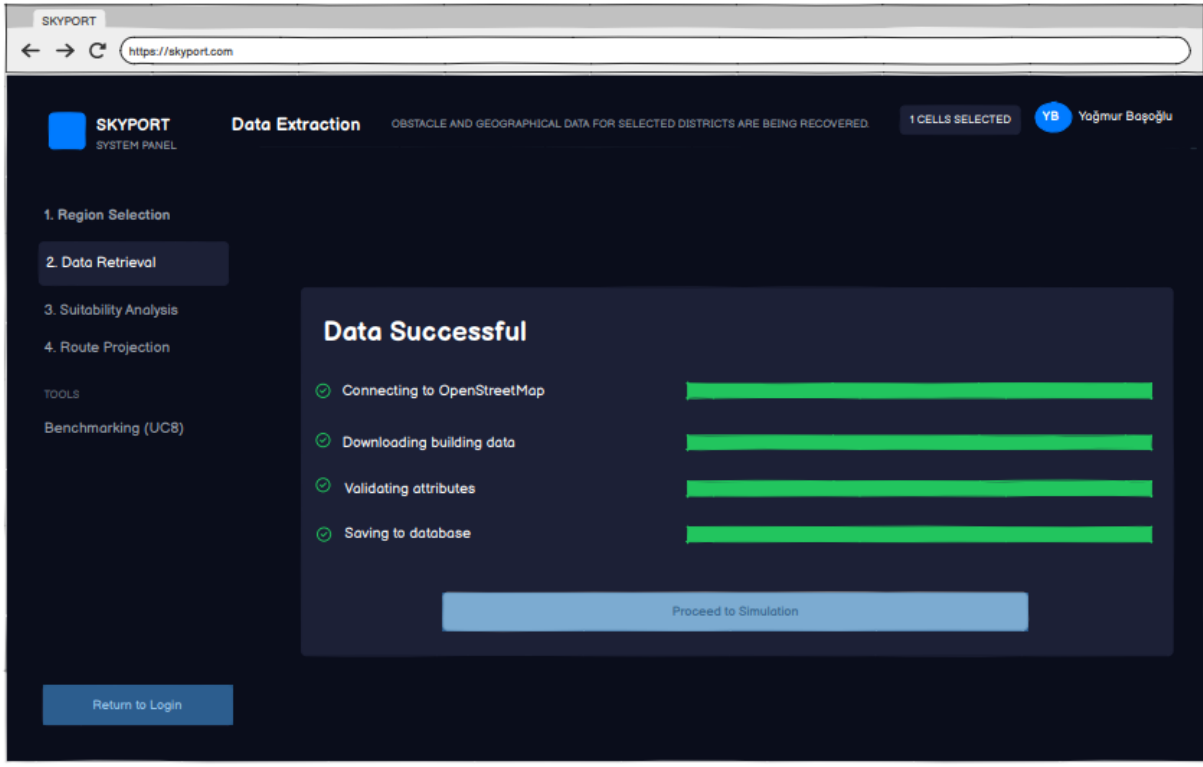


Figure 24

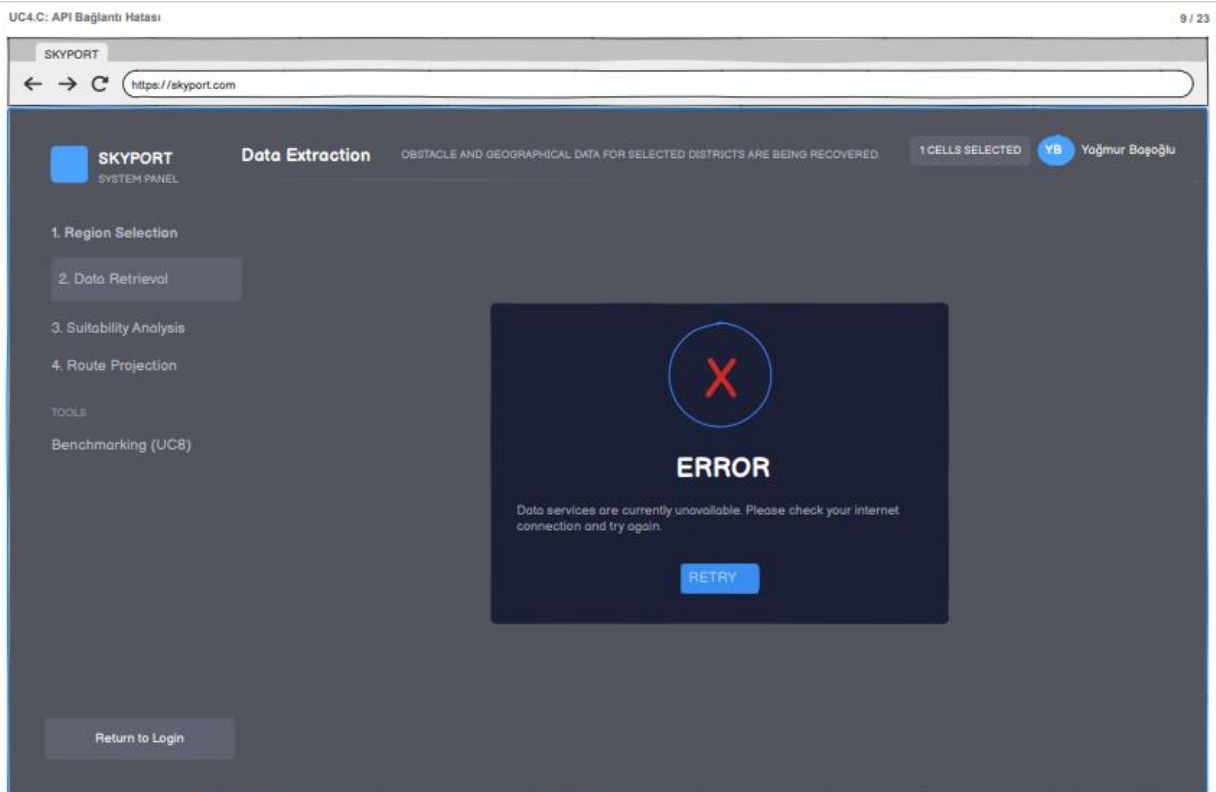


Figure 25

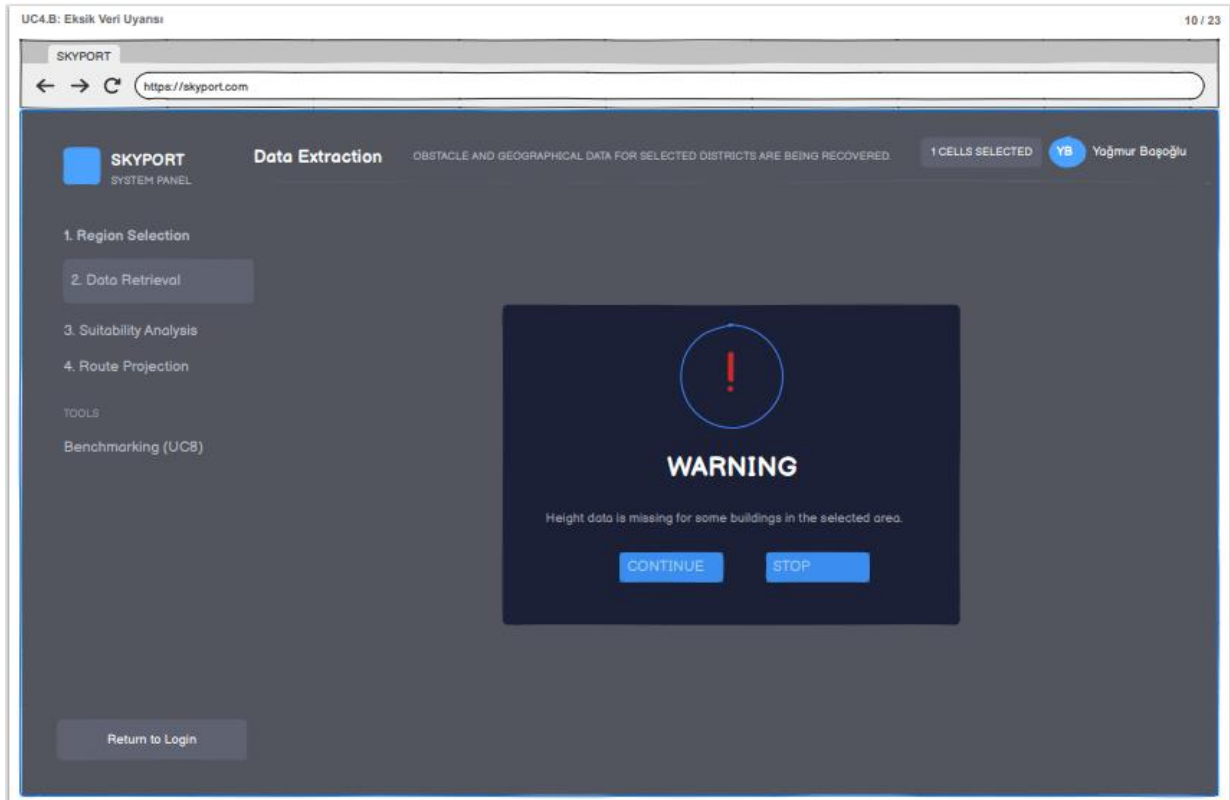


Figure 26

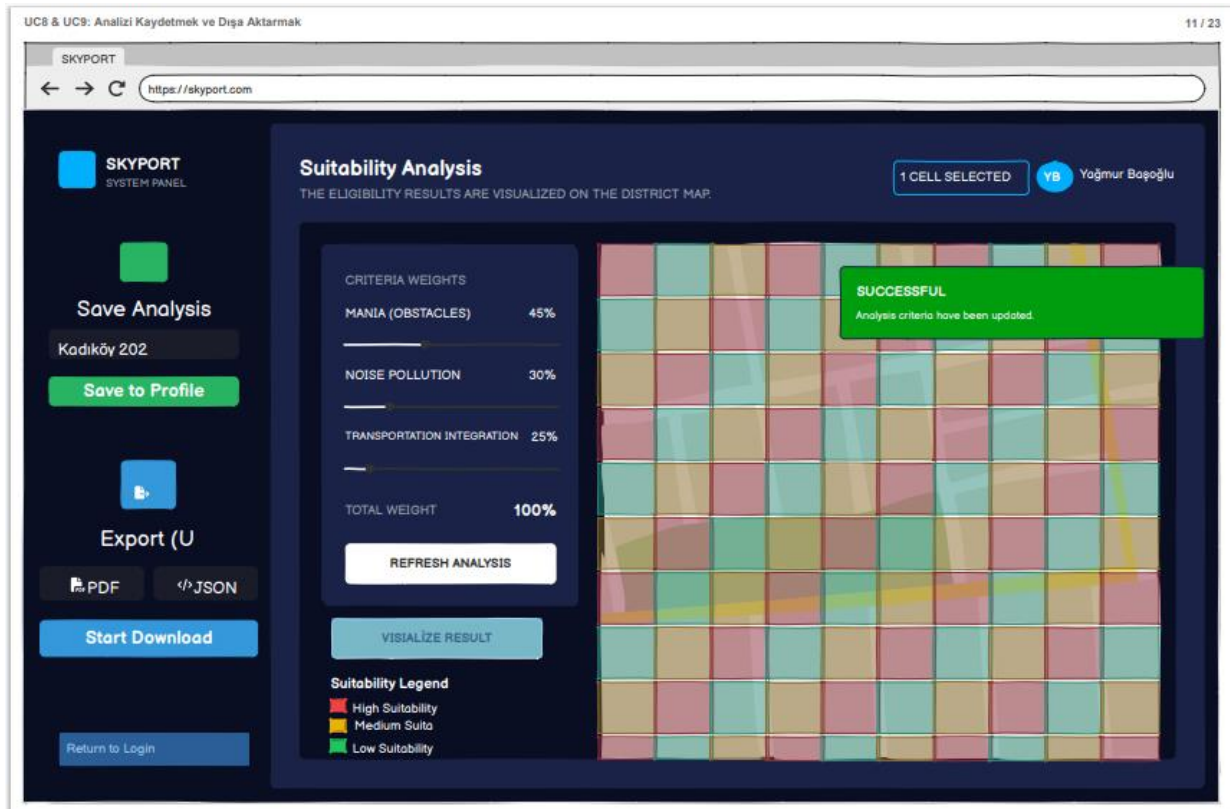


Figure 27

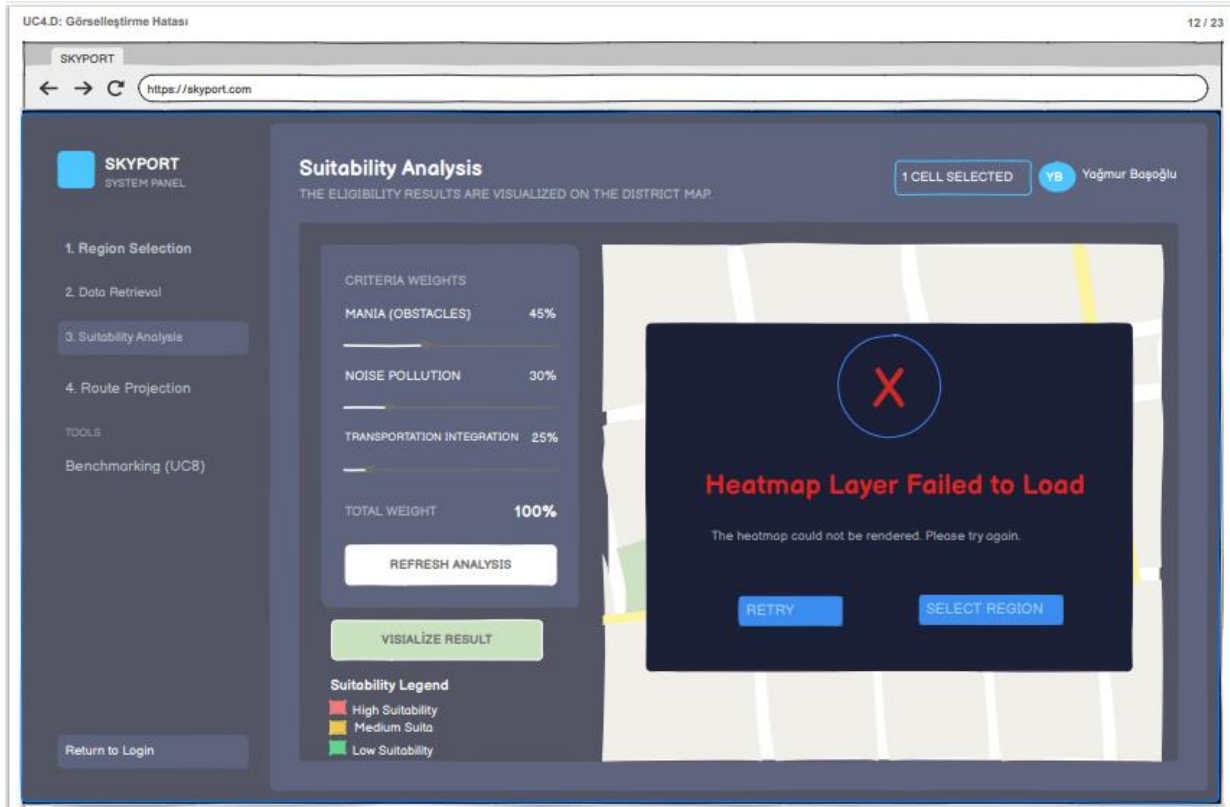


Figure 28

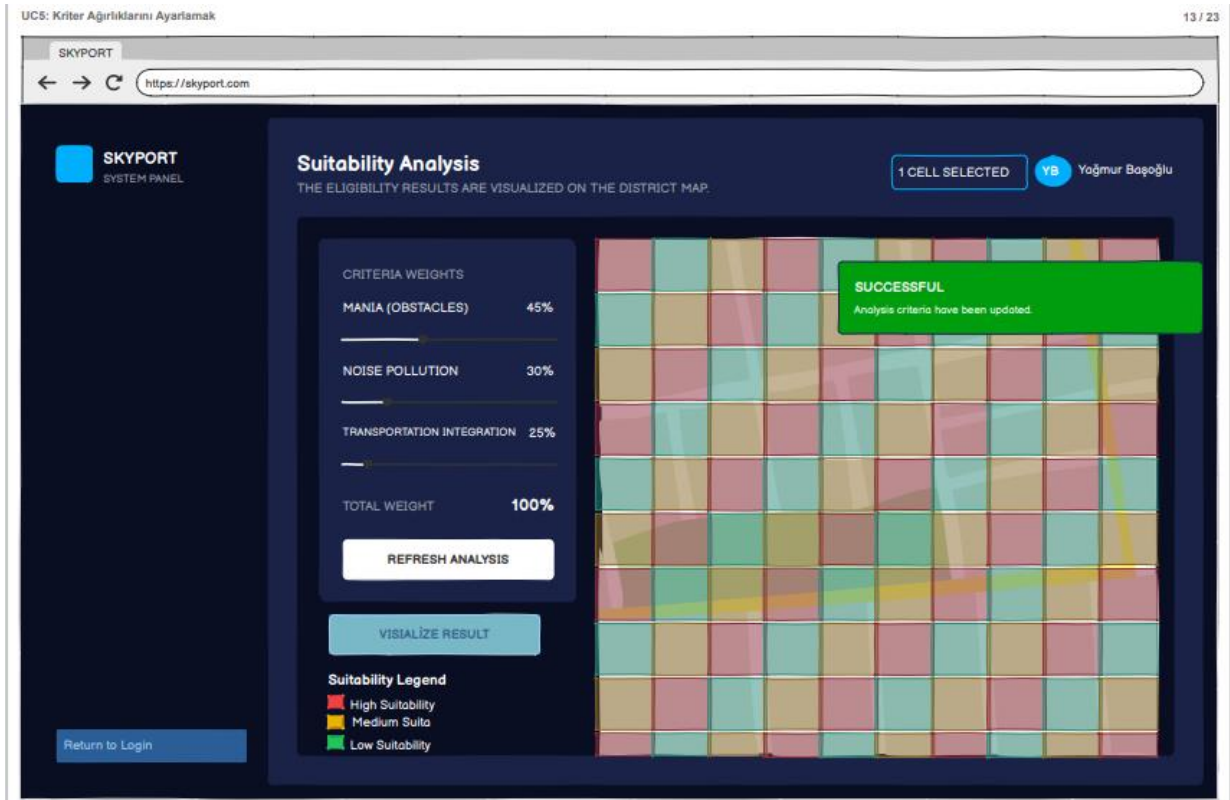


Figure 29

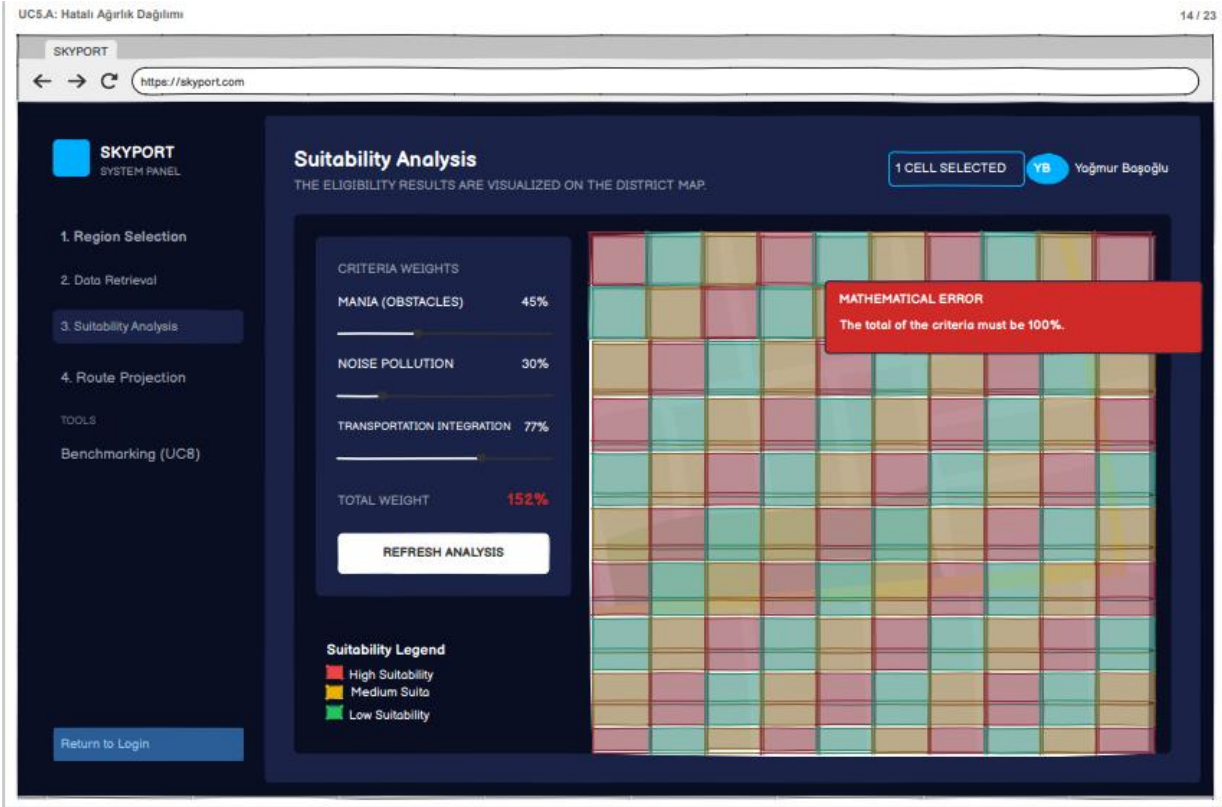


Figure 30

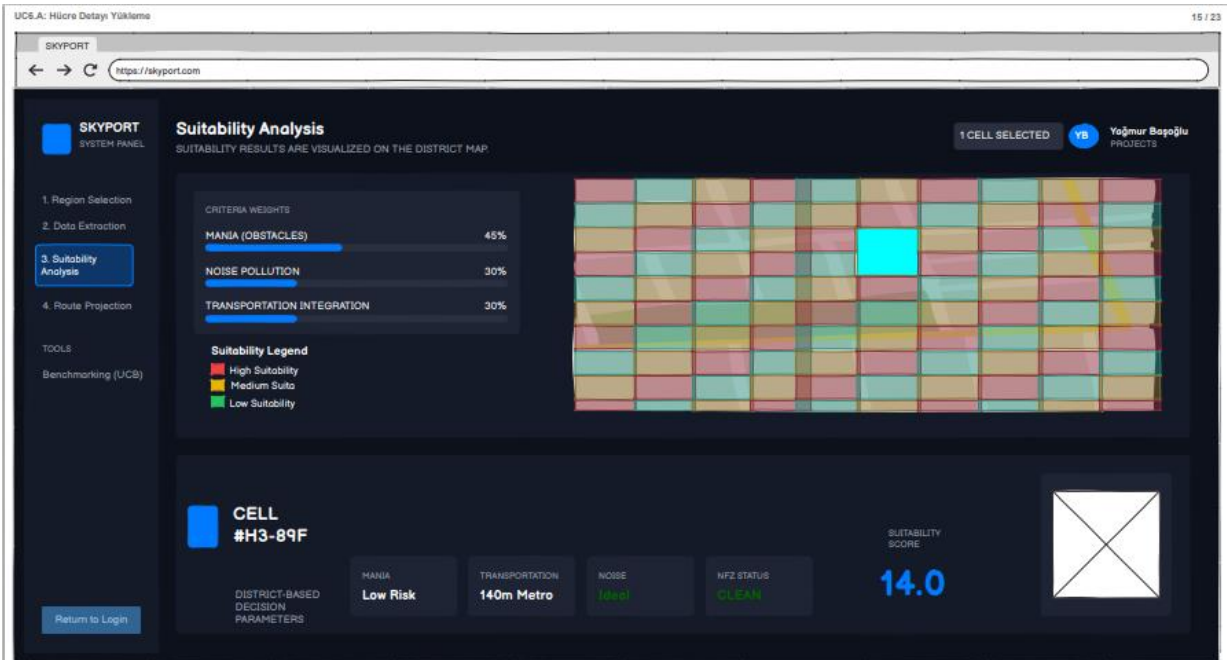


Figure 31

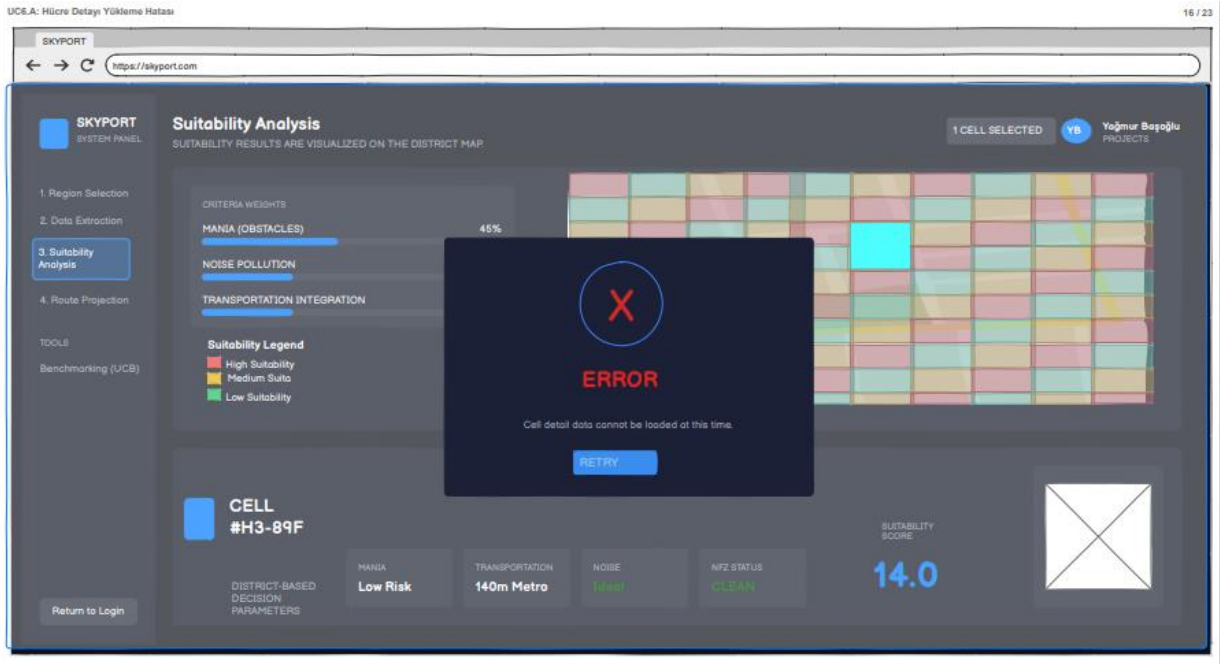


Figure 32

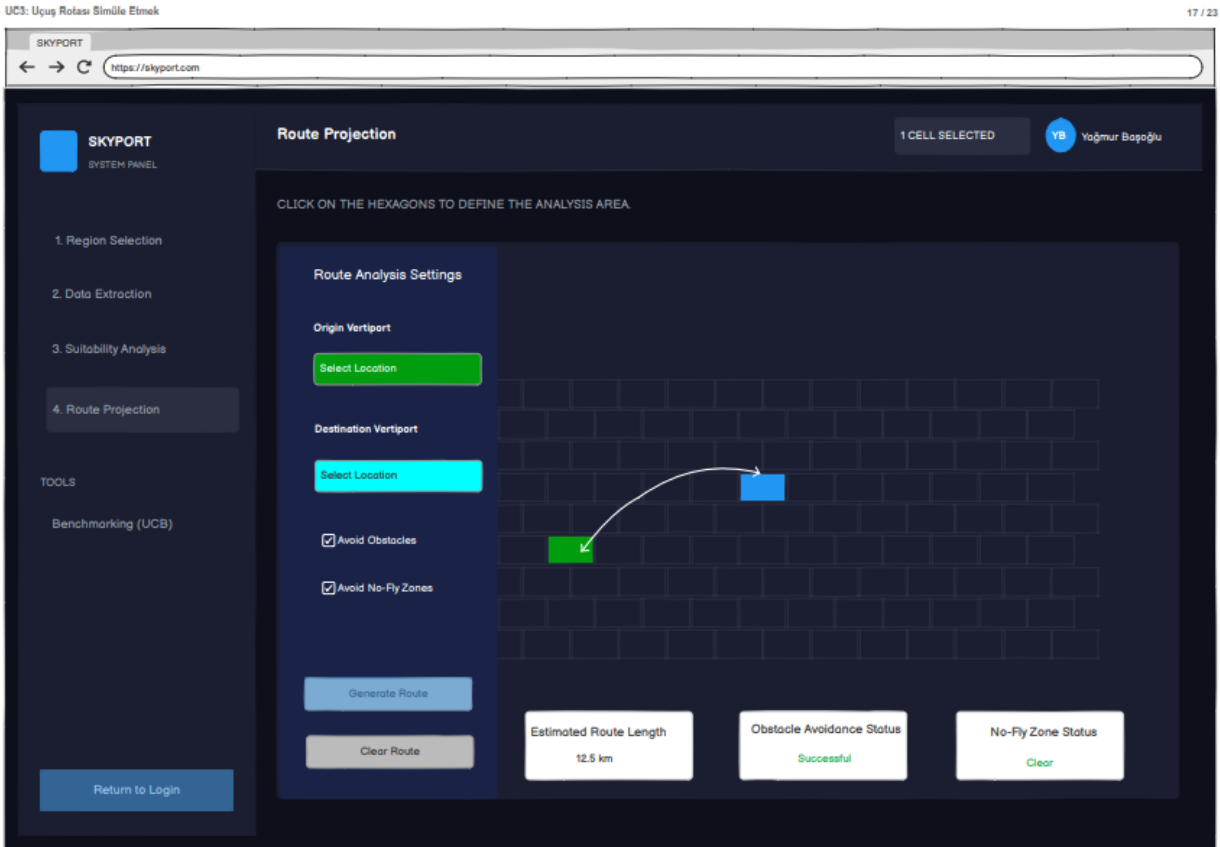


Figure 33

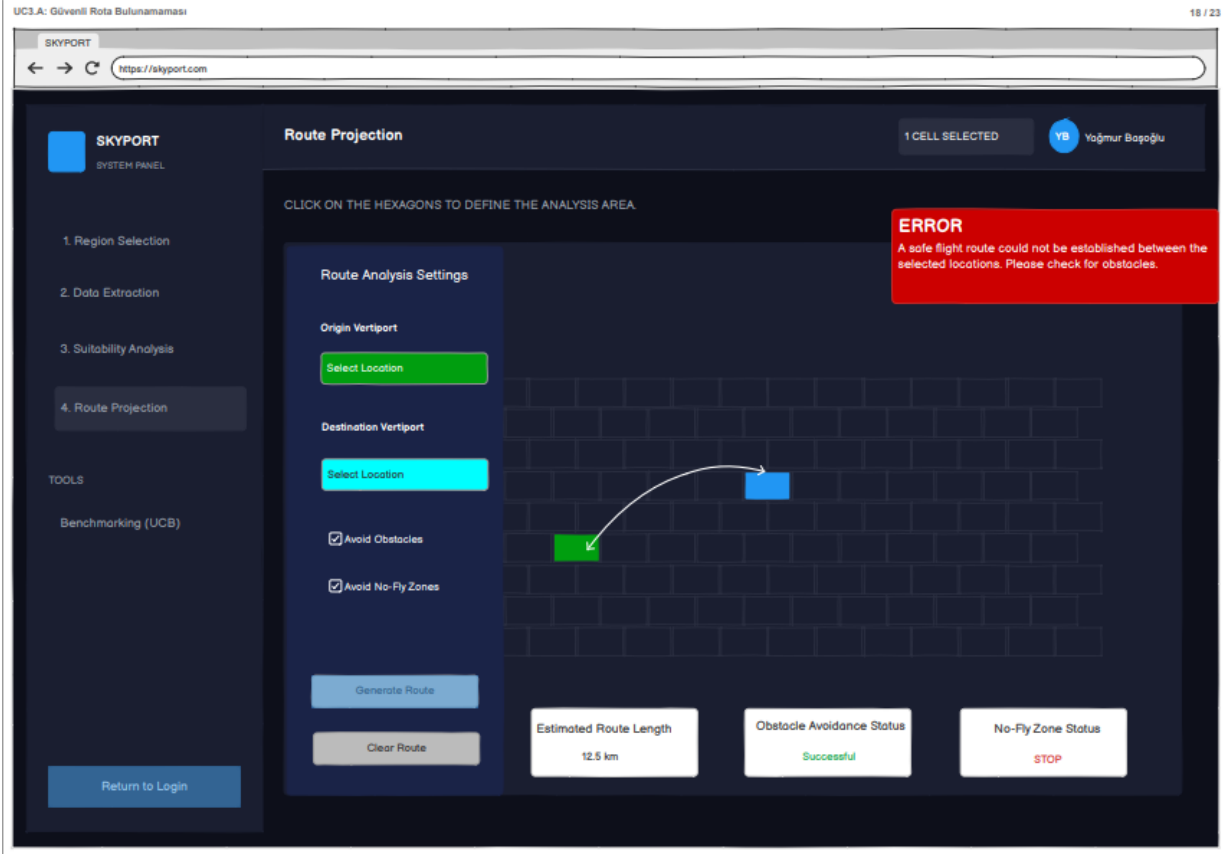


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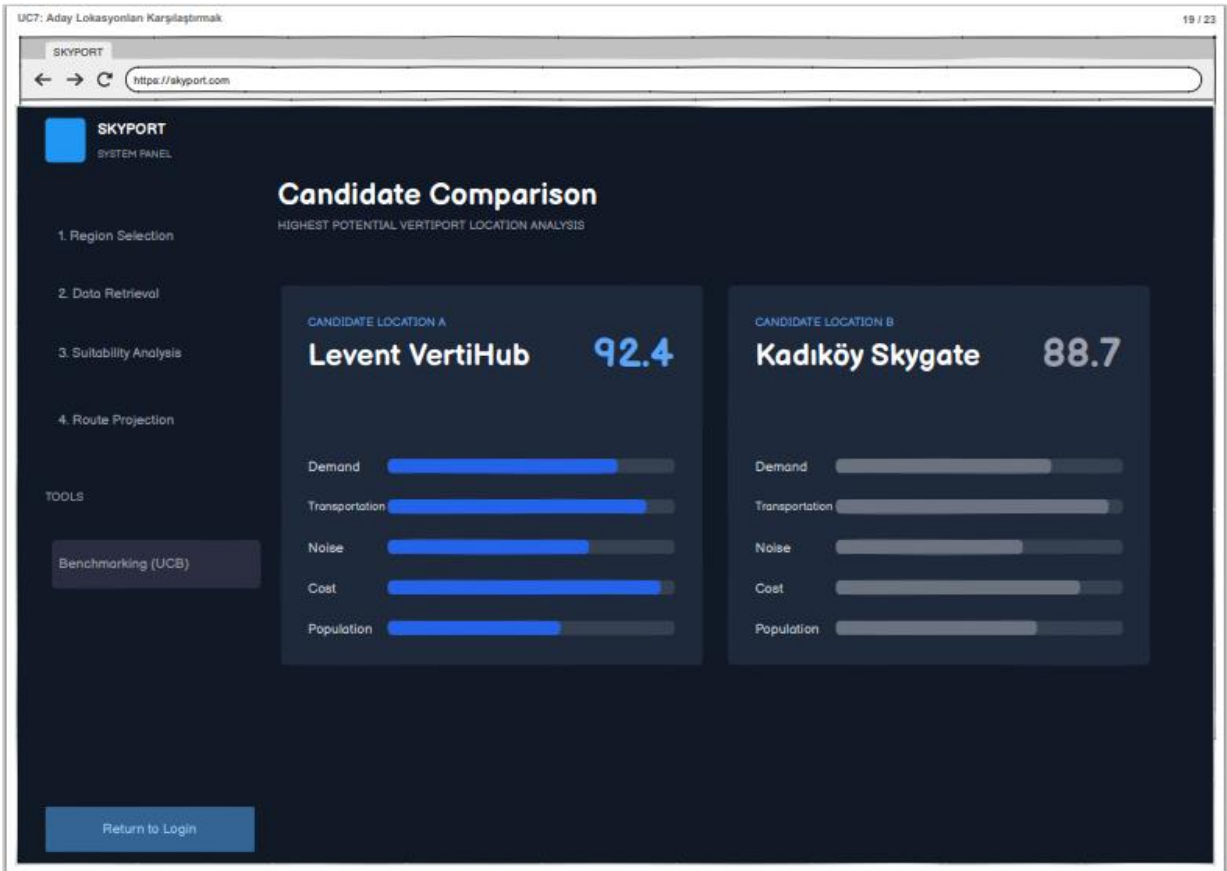


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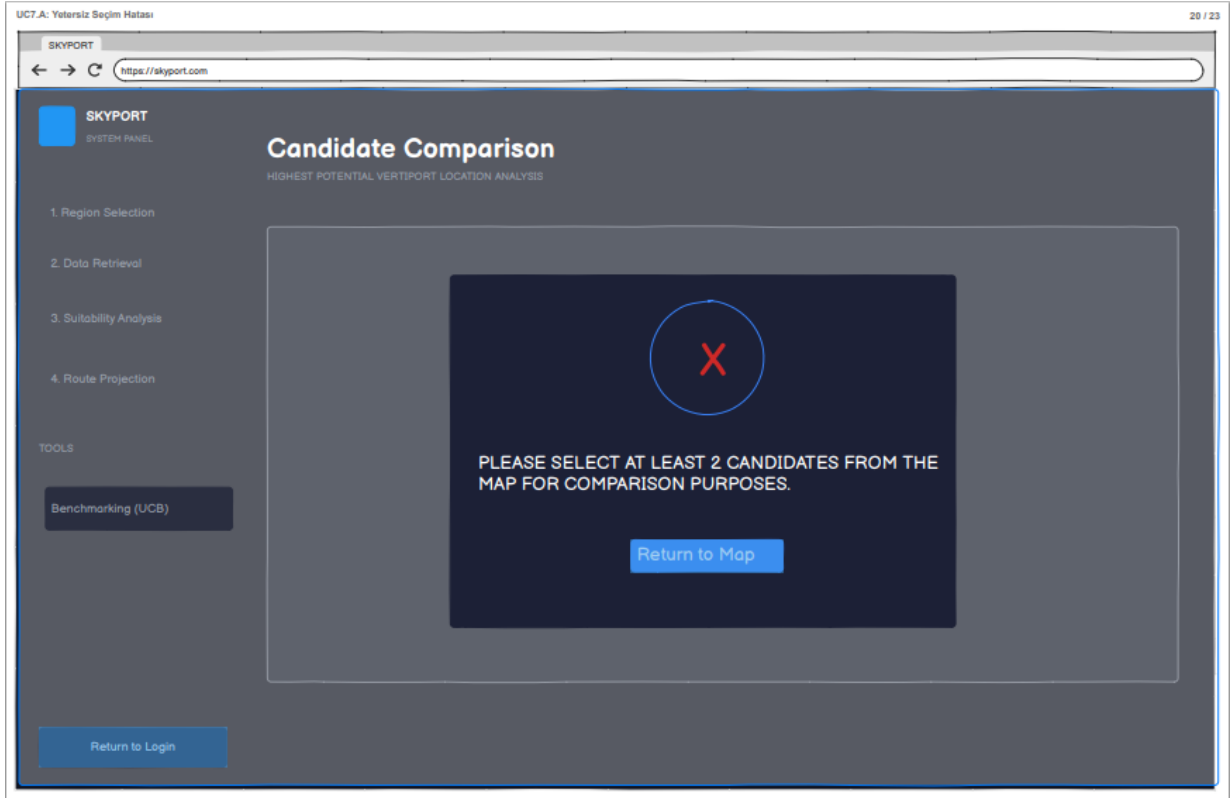


Figure 36

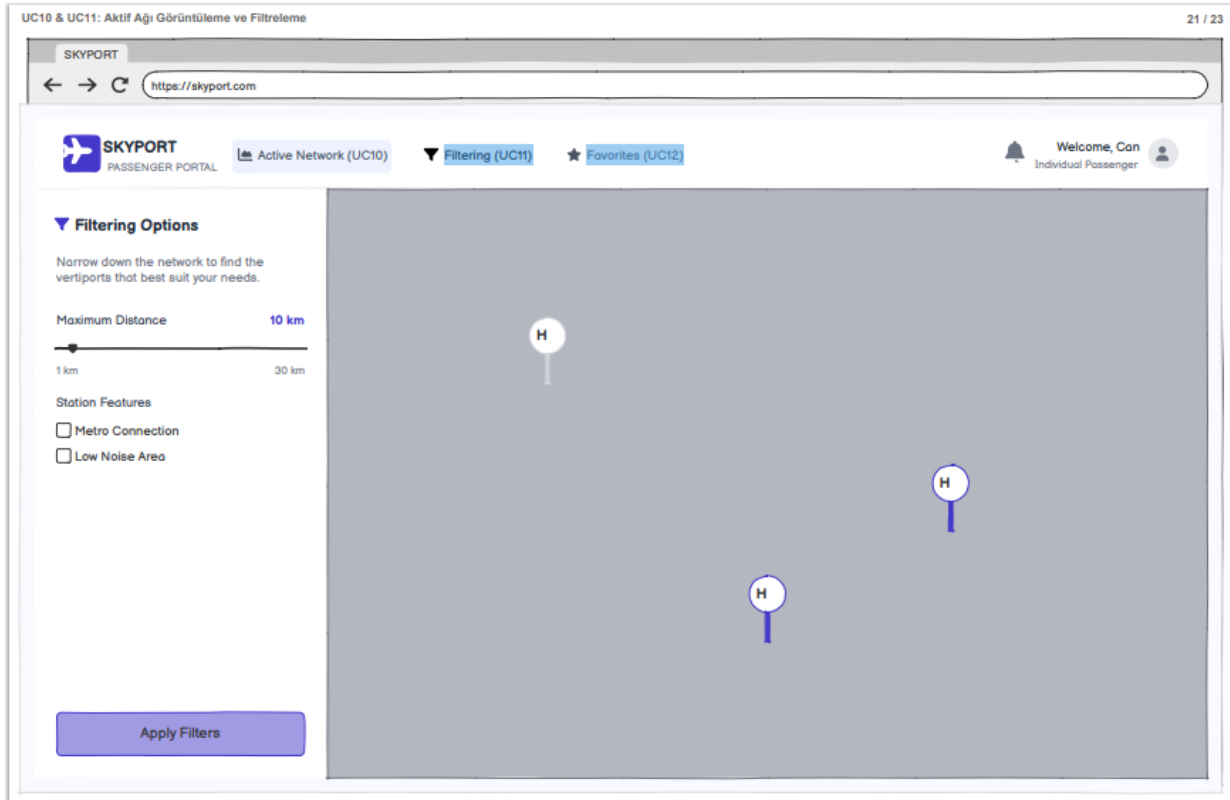


Figure 37

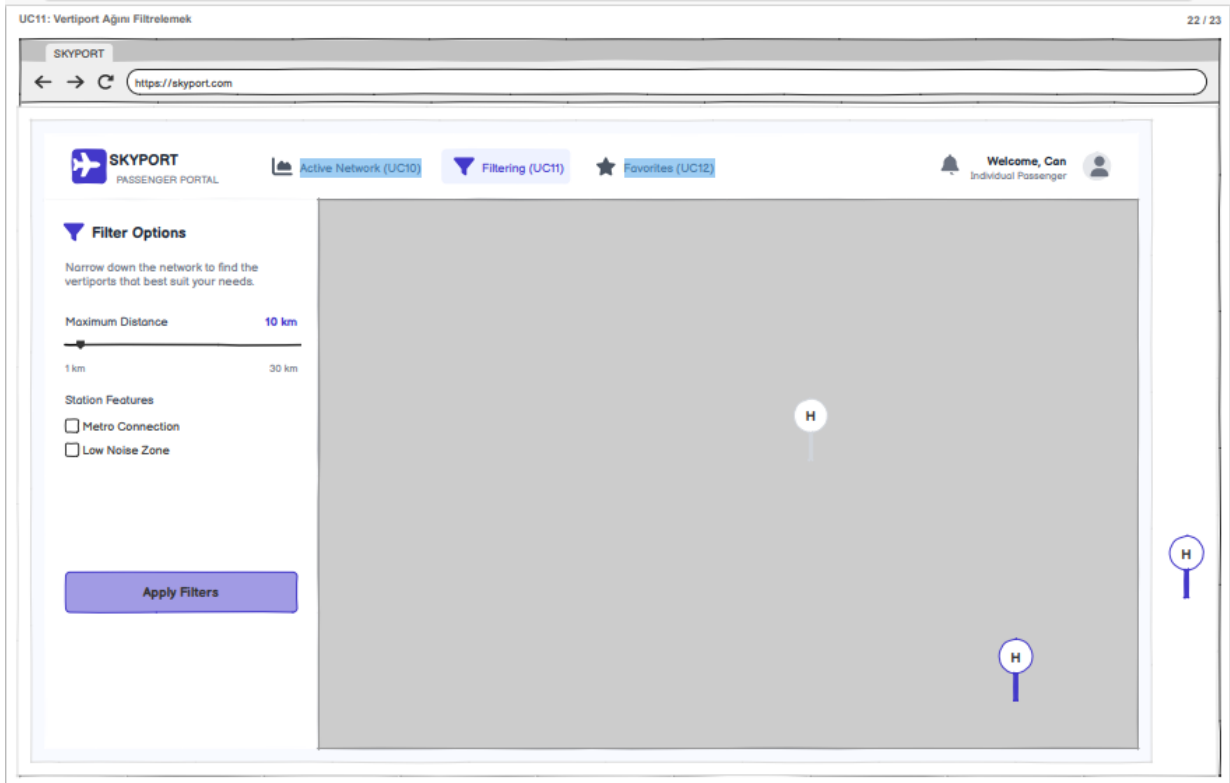


Figure 38

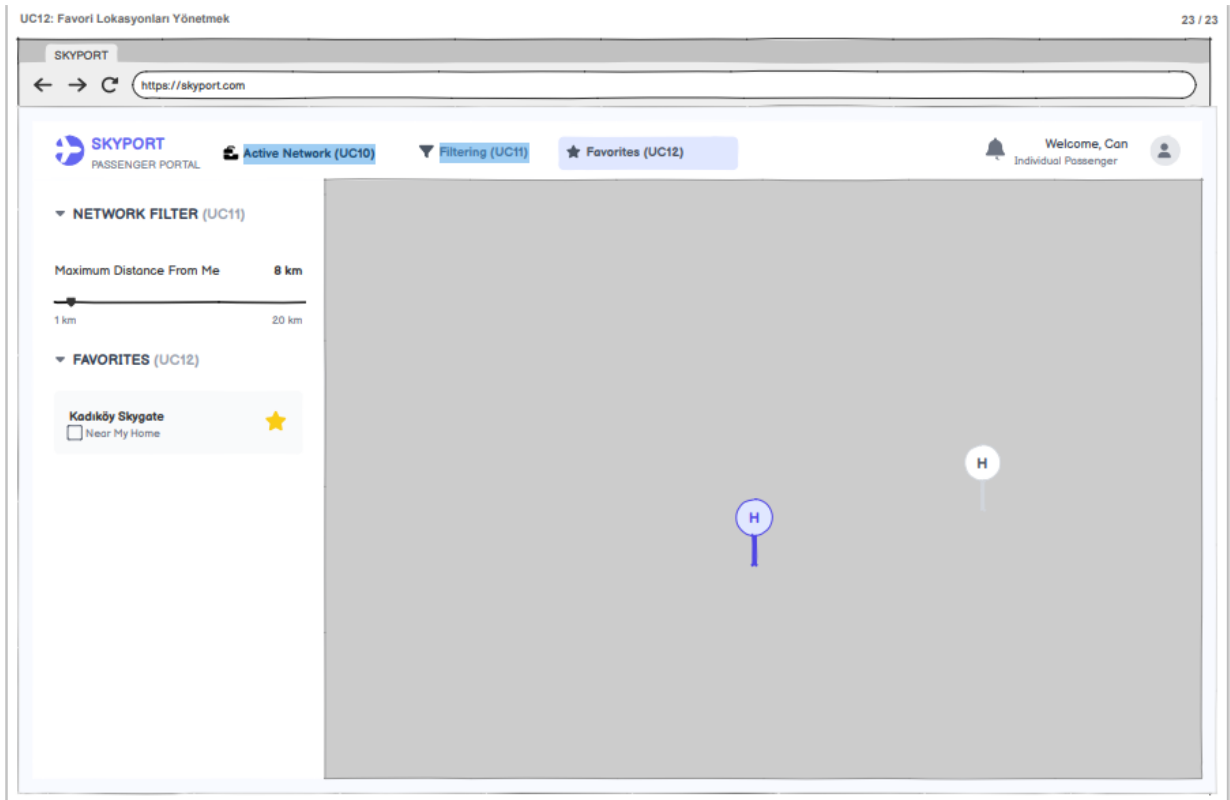


Figure 39

4. Other Analysis Elements

This section includes some other analyses concerning the development process of the SkyPort project. Risks that may occur during the project, as well as alternative solutions for these cases, are discussed. Moreover, the project's development plan and basic stages are briefly presented.

4.1 Risks and Alternatives

This section concerns the potential risks that may occur during the development process of the SkyPort project. More specifically, for each risk, the probability of the risk, the impact of the risk, and the alternative plan for the project, in case the risk occurs, are presented as follows:

Table 13

Risks	Likelihood	Effect on the project	B Plan Summary
Data Source Limitations: Inability to access data in the level of detail required (e.g., building heights) through OSM or open data sources.	Medium	The accuracy and coverage of the decision-making mechanism decrease	Default average values are given to the missing layers. The user is alerted to "missing data." The analysis process proceeds with partial reliability. Alternatively, Overture Maps API can be taken into account.
Complexity: AHP/TOPSIS mathematical models run slowly when processing thousands of H3 cells (large datasets).	Low	System response time is increased, along with user interface freezes (timeouts).	The H3 hexagonal grid resolution level used by the algorithm is reduced, or the massive analysis areas are divided into smaller "neighbourhood" based segments and processed asynchronously.
3D Visualization Challenges: Browser (client) based performance problems in the simultaneous rendering of building heights and routes in Mapbox GL JS.	Medium	The quality of the project is reduced.	The 3D layer is disabled, and the 2D heat map is prioritized.
Scope Creep: Delays in the development schedule due to the addition of features that were not part of the initial plan, like real-time flight tracking, weather, and payment gateways.	High	Project delivery deadlines may be delayed.	Other features include backlog, which means the scope of the current sprint is frozen, and prioritizing core functionality.
PostGIS/Database Performance: Query performance for large region queries becoming too slow.	Medium	Analysis/data retrieval time will be extended, and there will be a bottleneck risk.	Frequently used queries are cached; the maximum zone size for the analysis space is limited if necessary.

Export Errors: Errors that occur in converting complex map and analysis reports to PDF and GeoJSON formats.	Low	The reporting capabilities of B2B expert users will be limited.	A server-side reporting engine is used instead of browser-based libraries, or simplified text reports are provided.
Session and Data Security: Problems with database synchronization of passenger (B2C) favourites or expert (B2B) analysis records.	Low	User data loss will occur, and reliability will be impaired.	Standard and reliable authentication libraries (Firebase Auth, etc.) are used. Automatic backup scenarios are implemented to prevent data loss.

4.2 Project Plan

This section covers the development schedule and management strategy for the SkyPort project between February 2026 and June 2026. It is based on the technical requirements and key milestones identified during the project analysis phase.

Table 14: List of Work Packages

WP#	Work package title	Members involved
WP1	Requirements Elicitation and Analysis Processes	Yağmur Başoğlu, Nehir Karameşe
WP2	High-Level System Design & Database Modelling	Nehir Karameşe, Yağmur Başoğlu
WP3	Spatial Database Setup (PostGIS) and Data Extraction (OSMnx)	Yağmur Başoğlu
WP4	Frontend Framework Setup & Mapbox Base Layers	Nehir Karameşe
WP5	MCDM Algorithm Implementation & Python Backend Logic(Auth & Export)	Yağmur Başoğlu
WP6	Heatmap Visualization & B2B/B2C UI Components	Nehir Karameşe
WP7	API Integration, System Testing & Documentation	Yağmur Başoğlu, Nehir Karameşe

Table 15: Work Package 1

WP 1: Requirements Elicitation and Analysis Processes			
Start date: 02.03.2026 End date: 13.03.2026			
Leader:	Nehir Karameşe	Members involved:	Yağmur Başoğlu
Objectives: Before technical implementation of the project begins, the following steps are taken researching the existing literature, analysing aviation and urban planning standards, analysing			

<i>B2B/B2C user needs, and documenting all functional and technical requirements of the system.</i>
Tasks: Task 1.1 Literature Review: Researching the latest standards from EASA (European Union Aviation Safety Agency) and other relevant entities regarding the selection of Vertiport location and obstacles. Task 1.2 Technical Analysis and Data Source Research : Analysing the technical capabilities of the OpenStreetMap (OSM) data structure and the OSMnx library for Istanbul's 3D building data and transportation networks. Task 1.3 Requirements Definition (Functional & NFR): Detailed research regarding the functions that the system should fulfil. Task 1.4 Analysis Documentation : Combining the research with the technical analysis and creating the Proposal and Requirements Analysis Document (RAD/RDD).
Deliverables D1.1: Requirements Analysis Document (RAD/RDD)

Table 16: Work Package 2

WP 2: High-Level System Design & Database Modelling			
Start date: 16.03.2026 End date: 27.03.2026			
Leader:	Yağmur Başoğlu	Members involved:	Nehir Karameşe
Objectives: To design the high-level software architecture of the SkyPort system, design the PostGIS database schema used to store the spatial data (3D buildings, transportation networks, etc.), design the database schema (ERD) used to store the user profiles (B2B/B2C) and their favourites along with the spatial data, and define the interaction between the different components of the SkyPort system.			
Tasks: Task 2.1 System Architecture Design: Designing the 3-tier architecture with the React UI, Python-based backend with the choice of Flask or FastAPI, and the PostGIS database. Task 2.2 Database Modelling (ERD): Designing the database schema that includes the spatial data. Task 2.3 User Flow Creation: Designing the B2B and B2C interface flow design (wireframes). Task 2.3 Design Documentation: Documenting the design decisions and the architecture of the SkyPort system as a System Design Document (SDD).			
Deliverables D2.1: System Design Document (SDD)			

Table 17: Work Package 3

WP 3: Spatial Database Setup (PostGIS) and Data Extraction (OSMnx)			
Start date: 01.04.2026 End date: 30.04.2026			
Leader:	Yağmur Başoğlu	Members involved:	
Objectives: The objectives of the task are to configure the spatial extension of the PostgreSQL database using the PostGIS extension and develop Python/OSMnx scripts for extracting the building height, transportation networks, and land use information for the Istanbul region using the Open Street Map (OSM) data source.			

Tasks: Task 3.1 Database Installation: The task is about installing the PostgreSQL database and the PostGIS spatial extension, as well as the configuration of the spatial indexing settings (GIST). Task 3.2 Data Extraction Scripts: The task is about developing the Python/OSMnx scripts for extracting the building, road, and obstacle information for the Istanbul region. Task 3.3 Data Cleaning and Transfer: Cleaning the extracted raw data, performing coordinate system transformations, indexing to an H3 hexagonal grid structure, and transferring it to the database. Task 3.4 Data Validation: Validating the quality of the extracted data in the database and the completeness of the attributes of the data, particularly the height attribute. Deliverables D3.1: Structured and Data-Polluted Spatial Database. D3.2: Automatic Data Retrieval and Processing Module (Python Scripts).

Table 18: Work Package 4

WP 4: Frontend Framework Setup & Mapbox Base Layers			
Start date: 01.04.2026 End date: 30.04.2026			
Leader:	Nehir Karamese	Members involved:	
Objectives: The goal is to build the React.js framework, integrate the Mapbox GLJS map engine into the system, and create a basic map interface (base maps, navigation tools) for users to perform analysis.			
Tasks: Task 4.1 Frontend Setup: Creating the React.js project framework and configuring the necessary libraries (Mapbox, Axios, MUI, etc.). Task 4.2 Mapbox Integration: Defining the map API keys in the system and loading the base layer focused on Istanbul. Task 4.3 UI Component Design: Developing basic user interface controls such as zooming in, zooming out, and displaying coordinates on the map. Task 4.4 Area Selection Tools: Integrating the rectangular selection tool (Bounding Box) that will allow the user to define the analysis area on the map.			
Deliverables D4.1: Working Web Interface Framework. D4.2: Integrated Mapbox Map Module and Selection Tools.			

Table 19: Work Package 5

WP 5: MCDM Algorithm Implementation & Python Backend Logic (Auth & Export)			
Start date: 04.05.2026 End date: 22.05.2026			
Leader:	Yağmur Başoğlu	Members involved:	
Objectives: Code the Multi-Criteria Decision Making Algorithm develop a backend service using a Python backend to process criterion weights according to a normalization procedure, develop a simulation engine to compute fitness scores for all hexagonal cells in a hexagonal grid system, develop user sessions management (Auth), and develop an export function to export			

results from analysis in a report format.
Tasks: Task 5.1 Backend API Development : Setup a backend server using a Python backend server and define API endpoints that will interact with the frontend. Task 5.2 MCDM Algorithm Engine: Code the mathematical models for AHP (Analytic Hierarchy Process) and TOPSIS using a Python backend server to validate the models. Task 5.3 Database Integration: Code a database query layer to fetch geographic data required by the algorithm from a PostGIS database. Task 5.4 Scoring and GeoJSON Output: Developing the query layer that will retrieve the geographic data required by the algorithm from the PostGIS database. Task 5.5 Session management: Coding a module to manage user sessions (login and logout) and user profile management. Task 5.6 Export Output: Developing the report generation engine (GeoJSON/PDF Export).
Deliverables D5.1: MCDM Algorithm Module (Python) D5.2: Running Backend Simulation Service

Table 20: Work Package 6

WP 6: Heatmap Visualization & B2B/B2C UI Components			
Start date: 04.05.2026 End date: 22.05.2026			
Leader:	Nehir Karameşe	Members involved:	
Objectives: Display the results obtained from the analysis by the MCDM algorithm on the map as a dynamic heatmap on Mapbox, to add a set of control panels to allow the user to change the weights of the criteria, and to implement the cell detail review screen and filtering/favourite management for passengers.			
Tasks: Task 6.1 Heatmap Visualization : Adding calculated H3 score data in GeoJSON format to the map as color-coded (legend-based) layers using Mapbox GL JS. Task 6.2 Dynamic Control Panel: Developing "Slider" components by using React that allow the user to adjust criterion weights in real-time. Task 6.3 Detail Analysis Panel: Coding a side panel that opens when a cell is clicked and showing the score breakdowns (obstacles, transportation, etc.) of that location. Task 6.4 Comparison Module: Developing interface components that allow side-by-side comparison of multiple candidate locations. Task 6.5 Passenger Interface Panel: Implementation of passenger interface (B2C) filtering tools and favourite location management panel.			
Deliverables D6.1: Interactive Heat Map Layer D6.2: User Control and Analysis Panel			

Table 21: Work Package 7

WP 7: API Integration, System Testing & Documentation			
Start date: 25.05.2026 End date: 12.06.2026			
Leader:	<i>Yağmur Başoğlu</i>	Members involved:	<i>Nehir Karamişe</i>
Objectives: <i>To fully integrate the frontend and backend parts of the application with the API, conduct comprehensive testing of the application's reliability, and prepare the final thesis document for the project.</i>			
Tasks: Task 7.1 System Integration: <i>Ensuring communication between the React interface and Python backend services via REST API and optimizing data flow.</i> Task 7.2 Tests : <i>Performing unit and integration tests to debug and test software errors.</i> Task 7.3 Final Documentation and Thesis Writing: <i>Preparing and submitting the final Bachelor Thesis document by combining all development processes, architectural decisions, and test results.</i>			
Deliverables D7.1: <i>Fully Integrated and Tested SkyPort GDSS Software.</i> D7.2: <i>Final Bachelor Thesis & Poster Submission.</i>			

5. Glossary

Table 22: Glossary

AHP / TOPSIS	These are the mathematical techniques used by the algorithms of the Multi-Criteria Decision Making.
API (Application Programming Interface)	An interface that allows the interaction of two or more software.
B2B (Business-to-Business)	The part of the platform that deals with the analysis and management, specifically designed for experts (city planners, investors).
B2C (Business-to-Consumer)	The part of the platform that deals with the analysis and management specifically designed for the end-user (passenger).
Bounding Box	Rectangular boundaries defined by coordinates of the area to be analysed or from which the data is going to be extracted on the map.
eVTOL (Electric Vertical Take-off and Landing)	Next-generation electric-powered aircraft capable of vertical take-off and landing like helicopters (Flying taxis).
Export (	Downloading the analysis results and the map data that is generated within the system to the user's device.
GeoJSON	An open standard JSON data format used to encode geographic data structures.
GIS (Geographic Information System)	An information system that allows the collection, storage, analysis, and visualization of geographic data.
H3 / Hexagonal Grid	A geographic indexing system that allows the analysis of an area by dividing it into hexagonal cells. It is used in SkyPort for cell-based scoring of data.

Heatmap	The method of showing data values on a map and using colours, where red means unsuitable and green means suitable.
Mania	In aviation, any physical object that could compromise the flight safety of an aircraft.
Mapbox	A map-based visualization platform and API that enables the development of web-based map-based applications.
MCDM (Multi-Criteria Decision Making)	Multi-Criteria Decision Making, where multiple possibly conflicting, criteria are mathematically evaluated in order to achieve an objective.
OpenStreetMap (OSM)	An open-source geographic database created by volunteers worldwide. It includes road networks, building data, and various spatial data.
PostGIS	Software installed as an add-on to an open-source PostgreSQL database that allows the storage of Geographic Information System (GIS) objects.
Routing	A safe and optimum flight path that is simulated between two points, considering geographic constraints such as obstacles and restricted flight zones.
Urban Air Mobility(UAM)	This includes the transportation of passengers and/or cargo by air in urban environments, and it is the main area of interest of the SkyPort project.
Vertiport	Specific stations designed for UAM, where eVTOLs drop off and pick up passengers and are recharged.

6. References

Following list of references are relevant to this document, and they outline the technical, academic, and regulatory sources utilized while initiating the SkyPort project idea, analysing, and modelling its requirements.

- 1) Boeing, G. (2017). *OSMnx: New methods for acquiring, constructing, analysing, and visualizing complex street networks*. Computers, Environment and Urban Systems, 65, 126–139. <https://doi.org/10.1016/j.compenvurbsys.2017.05.004>
- 2) Işık University. (2026). *COMP/SOFT 4902/4912 Graduation Design Project Syllabus*. Işık University.
- 3) Lecture Notes, Işık University (2024). *Software Engineering Course Material (SOFT2101)*.
 - Course notes provided by the instructor for understanding the RAD structure, system modelling and diagrams.
- 4) Mapbox. (2026). *Mapbox GL JS Documentation for Web Maps*. Retrieved from: <https://docs.mapbox.com/mapbox-gl-js/>
- 5) Boeing, G. (2023). *OSMnx: Python for Street Networks*. Python Package Index (PyPI). Retrieved from: <https://pypi.org/project/osmnx/>
- 6) PostGIS Project Steering Committee. (2024). *PostGIS Spatial Database Documentation*. Retrieved from: <https://postgis.net/docs/>

- 7) Uber Technologies. (2024). *H3: Uber's Hexagonal Hierarchical Spatial Index*. Retrieved from:
<https://h3geo.org/>
- 8) United Nations. (2025). *The Sustainable Development Goals Report 2025*. Retrieved from:
<https://sdgs.un.org/goals>